Operating Leverage and Investment Adjustment in Times of Crisis: Evidence from COVID-19*

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Abstract

We study how operating leverage shapes firms' investment responses to unexpected revenue changes. We first develop a model in which past commitments that lower variable costs and current cost-reducing investments are substitutes. High-operating-leverage firms have already realized more cost savings through past commitments, leaving their remaining projects closer to the margin of profitability at the planned scale. When revenues fall unexpectedly, these firms therefore abandon a larger share of planned investment. We test this prediction using large-scale survey data from German firms collected during the COVID-19 pandemic. We measure operating leverage from the pass-through of revenue changes into profit changes, benchmarked against comparable firms experiencing the same revenue change. Consistent with the model, high-operating-leverage firms postpone and cancel 19.6% and 41.3% more of their planned investment, respectively, than low-operating-leverage firms, and they are less likely to plan investment over the subsequent two years. Additional analyses indicate that financial constraints do not fully explain these differences. Overall, the results show that operating leverage shapes investment beyond its effect on firms' ability to finance it.

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1 Introduction

Operating leverage is studied primarily as a determinant of risk. A rigid cost structure amplifies the effect of revenue shocks on profit, thereby increasing a firm's systematic risk and shaping its financial policy (Lev, 1974; Mandelker and Rhee, 1984; Novy-Marx, 2011; Chen et al., 2019). In bad times, this amplification means that firms with higher operating leverage experience larger profit declines, which may leave them with fewer internal resources to fund investment. This is consistent with evidence from the COVID-19 pandemic that firms entering the crisis with greater financial flexibility were better able to withstand it (Ding et al., 2021; Fahlenbrach et al., 2021). In this view, operating leverage affects a firm's *ability* to invest. At the same time, a firm's cost structure may also affect the attractiveness of investment itself, because it shapes which projects remain worthwhile once demand falls. Operating leverage may therefore affect not only a firm's ability, but also its *desire* to invest. This paper asks whether and how operating leverage shapes firms' investment responses to a demand shock, and whether any such responses can be fully explained by financial constraints.

The COVID-19 pandemic provides a useful setting to study this question because it exposed firms to a large and unexpected shock to revenues. Many firms responded by postponing or canceling investment, prompting governments and central banks to support liquidity and credit provision (Board of Governors of the Federal Reserve System, 2020; Koalitionsausschuss, 2020). Such measures can preserve firms' ability to finance investment, but they do not ensure that projects planned before the shock remain profitable afterward. When demand falls, projects whose returns depend on the volume over which they are used may no longer be profitable even if financing remains available. Understanding this distinction matters for policy because measures that ease financing constraints need not preserve investment if firms no longer find the underlying projects attractive.

We formalize this idea in a model of firms that differ in operating leverage and can invest to reduce variable costs. Past commitments and current cost-reducing investments are substitutes as firms with more rigid structures have already realized more of the available cost savings through prior commitments, leaving the remaining investment projects closer to the margin of profitability. When demand falls, the return to these projects declines because their cost savings apply to fewer units. The model therefore predicts that firms with more rigid cost structures revise a larger share of their planned investment and that the extent of revision increases with operating leverage. Importantly, these predictions arise without financing frictions: investment falls because fewer projects remain worthwhile at the lower level of demand, not because firms lack financing resources.

Next, we test the model's predictions using large-scale survey data from the German Business Panel (GBP), which samples from the universe of legal entities in the German Statistical Business Register. Our main sample contains 6,786 firms reporting a change in revenue following the outbreak of the COVID-19 pandemic, including 4,873 firms reporting a decline. The data were collected between July and October 2020, while the economic effects of the shock were still unfolding. A key advantage of the survey is that it records firms' revisions to investment plans formed before the crisis. This margin is generally not recoverable from archival data because an investment that was planned and subsequently postponed or canceled is observationally identical to one that was never intended in the first place. The GBP instead records the share of planned investment that firms postponed or canceled, allowing us to study revisions to investment intentions directly.

We measure operating leverage by comparing each firm's realized profit response during the crisis with the response predicted by its revenue change and firm characteristics. Specifically, we regress the reported change in profit on the reported change in revenue, industry, firm size, and survey month. For firms experiencing a revenue decline, we classify a firm as having high operating leverage when its profit falls by more than predicted, consistent with a cost structure that adjusts less readily in the short run. Although this measure does not rely directly on cost data, a follow-up wave of the GBP allows us to validate its interpretation. Firms classified as having high operating leverage report greater difficulty reducing costs in the short run ($p < 0.01$), and the obstacles that distinguish the two groups are concentrated in commitments entered into before the crisis.

In a second step, we use this measure of operating leverage to study firms' investment adjustments during the crisis. We relate the operating leverage classification to the share of the pre-crisis investment program that firms postpone or cancel and to their investment plans over the following two years. The regressions control for industry and firm size. To separate operating leverage from differences in the severity of the shock, we additionally include a separate fixed effect for each reported revenue change. Furthermore, we control for firms' expectations of GDP growth over the following two years, mitigating the concern that differences in our operating leverage classification and subsequent investment revisions reflect firms' expectations about the economic recovery. Identification therefore comes from differences in operating leverage among firms reporting the same revenue decline, conditional on industry, firm size, and expectations about future economic conditions.

The regression results support the model's main prediction. Firms classified as having high operating leverage revise a larger share of the investment they had planned before the crisis. They

postpone 4.8 and cancel 5.9 percentage points more of their pre-crisis investment program than firms with low operating leverage, corresponding to 19.6 and 41.3 percent of the respective means for the low operating leverage group (both $p < 0.01$). Furthermore, they are 3.6 and 4.6 percentage points less likely to plan investment over the next twelve and the subsequent twelve to twenty-four months (both $p < 0.01$). The revision of existing investment plans also increases with the degree of operating leverage. Specifically, replacing the binary classification with the standardized distance between a firm's realized and predicted profit change, a one-standard-deviation increase in operating leverage raises the postponed share by 2.9 percentage points and the canceled share by 3.0 percentage points. We do not find a comparable gradient for whether firms plan to invest at all. The results are robust to alternative specifications, samples, and estimators. They also remain similar when we replace the first-stage classification with a simple pass-through measure based on the ratio of firms' profit and revenue changes.

In our remaining analyses, we provide evidence that is difficult to reconcile with an explanation in which the investment differential is driven entirely by financial constraints. First, the association between operating leverage and investment adjustments persists when we control for contemporaneous changes in liquidity, employment, and liabilities. This is a demanding specification because these variables are themselves affected by the shock. The coefficients on the postponed and canceled shares attenuate to 2.8 and 4.1 percentage points, respectively, but both remain significant at the one percent level. Second, the association does not become stronger as financing constraints become more binding. The GBP data allow us to observe which firms received government support and whether they report that they would have survived the crisis without it. Financial constraints are plausibly most binding among the 1,750 firms that state they would not have survived without support. The investment differential between firms with high and low operating leverage does not become stronger in this group, and the estimates are statistically indistinguishable from those for firms reporting less reliance on government support. Third, among the 1,913 firms whose revenues rose, the association tends to reverse, a pattern that a mechanism operating through scarce internal resources does not naturally predict. Taken together, these results suggest that financial constraints may contribute to firms' investment responses but do not fully account for the association with operating leverage.

The contribution of this paper is threefold. First, we show theoretically how an inherited cost structure shapes firms' investment revisions after an unexpected revenue shock. Research on strategic and cost-reducing investment studies how investment commitments affect subsequent

competition, production costs, and prices (Spence, 1977; Dixit, 1980; Brander and Spencer, 1983; Fudenberg and Tirole, 1984; d'Aspremont and Jacquemin, 1988; Hattori and Tanaka, 2016). In these models, investment typically precedes and helps determine the firm's later cost position. We study the reverse sequence: firms enter the shock with asymmetric cost structures already in place and then revise investment once demand changes. This creates a substitution between past cost commitments and current cost-reducing investment, so firms with more rigid structures cut back a larger share of their planned investment. Related work in finance studies how competition and market leadership affect risk and expected returns (Aguerrevere, 2009; Carlson et al., 2014), while research on investment under uncertainty shows how uncertainty and irreversibility shape investment responses (Guiso and Parigi, 1999; Bloom et al., 2007). We add a real-decision channel in which firms facing the same revenue shock respond differently because their inherited cost structures leave different projects worth undertaking. Finally, accounting research studies how firms arrive at different cost structures (Kallapur and Eldenburg, 2005; Banker et al., 2014). We take those structures as given and examine their consequences for subsequent investment decisions.

Our second contribution is to develop a survey-based measure of operating leverage that can be applied when detailed cost data or long accounting histories are unavailable. Existing measures typically infer operating leverage either from the composition of firms' costs (Lev, 1974; Novy-Marx, 2011; Aboody et al., 2018) or from the time-series sensitivity of profits to sales (Mandelker and Rhee, 1984; O'Brien and Vanderheiden, 1987). Both approaches are difficult, if not impossible, to apply to privately held firms, for which detailed cost information and repeated financial statements are often unavailable (Kallapur and Eldenburg, 2005; Minnis and Shroff, 2017). We instead exploit a common shock observed across many firms and infer operating leverage from how strongly each firm's profit changes relative to its revenue change and to comparable firms facing the same shock. A separate survey wave supports this interpretation, as firms classified as having higher operating leverage subsequently report greater difficulty reducing costs in the short run. More broadly, the approach illustrates how survey data can recover economically relevant firm characteristics that are difficult to observe in archival data (Graham and Harvey, 2001; Graham, 2022).

Our third contribution is to document a real investment consequence of operating leverage that goes beyond its effects on risk and financing. Prior research shows that operating leverage affects expected returns, profitability, and capital structure (Chen et al., 2019; Donangelo et al., 2019), and that these effects depend on firms' ability to adjust their scale or other operating inputs (Reinartz and Schmid, 2016; Serfling, 2016; Gu et al., 2018). Bhojraj et al. (2021) show that the

consequences of a firm's cost structure depend on the macroeconomic conditions it expects, and that credit markets reflect this only incompletely. Closest to our setting, Barry et al. (2022) show that financial flexibility supports employment and investment more strongly when fixed costs are high, highlighting how cost structure shapes the value of financing capacity during a crisis. We examine the complementary margin. Holding constant the severity of the shock and accounting for financing conditions, firms with more rigid cost structures revise a larger share of planned investment, consistent with operating leverage changing which projects remain worth undertaking. This evidence links firms' cost structures to subsequent real decisions, complementing the cost-behavior literature that studies how committed resources adjust when activity changes (Anderson et al., 2003; Banker et al., 2013; Banker and Byzalov, 2014).

The remainder of the paper proceeds as follows. Section 2 develops the model and derives the two predictions we test. Section 3 describes our data, derived from the GBP, the construction of our measure of operating leverage, and the specification we estimate. Section 4 reports the results, and Section 5 concludes.

2 A Model of Operating Leverage, Investments, and Demand Shocks

We develop a stylized model to illustrate how a firm's cost structure shapes its investment response to a revenue shock. The model has three main ingredients. First, firms differ in their cost structure, which is the result of past decisions and cannot be adjusted in the short run. Second, firms can undertake cost-reducing investments. Third, demand changes unexpectedly, as it did during the COVID-19 pandemic, and firms may revise their output and investment decisions in response.

We embed these ingredients in a market with a leader and a follower. We assume the leader to be the firm with the more rigid cost structure: it has committed to higher fixed costs in exchange for lower unit variable costs, that is, to more fixed and less variable cost in the sense of Lev (1974). Throughout, we refer to it as the firm with higher operating leverage. The two features have a common source. A firm leads its market because it has made commitments that its rivals must take as given, such as production capacity, technology, and long-term supply and employment contracts, and it is precisely the irreversibility of these commitments that makes them strategically effective (Spence, 1977; Dixit, 1980). The same commitments fix a large part of the firm's cost base and lower the cost of each additional unit it sells. Operating leverage is thus the cost-structure counterpart of the leader's strategic position.

2.1 Setup

Market. Two firms, a leader (L) and a follower (F), sell a homogeneous product. The inverse demand function is

$$p = d - (q_L + q_F), \tag{1}$$

where q_i denotes the output of firm $i \in \{L, F\}$ and $d > 0$ measures the level of demand. A demand shock is a change in d.[1]

Cost structure. Each firm i bears a committed fixed cost $K_i \geq 0$ and produces at a baseline unit variable cost $c_i > 0$. Both result from past decisions on capacity, technology, and contracting, such as leases, long-term employment contracts, or automation, which firms make in view of the demand they expect (Banker et al., 2014). Neither can be adjusted in the short run. In line with the leader's more rigid cost structure, we assume

$$K_L > K_F, \qquad c_L < c_F \leq 1. \tag{2}$$

The bound on c_F is a regularity condition that keeps the follower's problem well behaved (Lemma 2). Committed costs do not affect any marginal decision; they matter only for the level of profits.

Investment. Each firm can invest an amount $I_i \geq 0$ in cost reduction, for example in process improvements or automation. Investment lowers the unit variable cost to

$$\kappa_i(I_i) = c_i \, e^{-I_i}. \tag{3}$$

Each additional unit of investment reduces the unit variable cost by the same percentage, so the payoff from investment scales with the variable cost base, $c_i q_i$, to which the cost reduction applies. Accordingly, firm i's profit amounts to

$$\pi_i = [p - \kappa_i(I_i)] \, q_i - I_i - K_i. \tag{4}$$

Timing. Demand d is common knowledge. In stage 1, the leader chooses its output q_L and investment I_L. In stage 2, the follower observes q_L and chooses q_F and I_F.[2] We solve for the

[1] The unit slope of the inverse demand function is a normalization. For $p = d - m(q_L + q_F)$ with $m > 0$, measuring quantities in units of $1/\sqrt{m}$ and prices in units of $\sqrt{m}$ leaves all profits unchanged and yields (1). All parameter conditions below are therefore conditions on $d/\sqrt{m}$ and $c_i/\sqrt{m}$.

[2] Whether the follower also observes I_L is immaterial, because I_L does not affect the follower's payoff.

subgame-perfect equilibrium. Within each stage, a firm chooses output and investment jointly, which captures that, in a sudden crisis, firms revise production plans and investment programs at the same time. Investment therefore serves to minimize the cost of the chosen output rather than to induce the rival to produce less, as in models in which investment precedes output competition and commits the investing firm to a larger output (e.g., Fudenberg and Tirole, 1984).

We interpret I_i as the firm's investment program for the period: chosen in view of the demand the firm expected, and not yet carried out when the shock arrives. Because the decline from d_0 to d_1 is unanticipated, the program the firm had set is a plan it genuinely held, and the cost structure it committed to earlier cannot have been chosen in anticipation of the shock. Comparing equilibrium investment at a pre-shock demand level d_0 with that at a post-shock level $d_1 < d_0$ therefore describes how a firm revises its planned investment when demand unexpectedly falls.

2.2 Cost-Minimizing Investment

Investment affects only the investing firm's own unit cost. Hence, for any output it produces, each firm chooses investment to minimize its total variable and investment cost,

$$C_i(q) \equiv \min_{I \geq 0} \left\{ c_i e^{-I} q + I \right\}. \tag{5}$$

Lemma 1 (Investment rule)**.** *For every output level $q \geq 0$:*

(a) The unique cost-minimizing investment is $I_i^*(0) = 0$ and $I_i^*(q) = \max\{0,\ \ln(c_i q)\}$ for $q > 0$. Firm i invests if and only if $c_i q > 1$.

(b) The minimized cost is $C_i(q) = c_i q$ if $c_i q \leq 1$ and $C_i(q) = 1 + \ln(c_i q)$ if $c_i q > 1$. C_i is continuously differentiable, strictly increasing, and concave, with marginal cost $C_i'(q) = \min\{c_i,\ 1/q\}$ for $q > 0$ and $C_i'(0) = c_i$.

(c) If $c_i q > 1$, then $\partial I_i^* / \partial c_i = 1/c_i > 0$ and $\partial \ln I_i^* / \partial \ln q = 1/I_i^*$.

Proof. See Appendix A. □

A firm invests only if its variable cost base, $c_i q$, is large enough to make cost reduction worthwhile. Moreover, cost rigidity and investment are substitutes: at a given output, a firm that already produces at a low unit variable cost has less to gain from further cost reduction and runs a smaller investment program (part (c)). Because the elasticity of investment with respect to output equals

$1/I_i^*$, a smaller program reacts more strongly, in relative terms, to a given change in output. This property drives our main result.

2.3 Equilibrium

Follower. Given the leader's output q_L, let $a \equiv d - q_L$ denote the intercept of the follower's residual demand. By Lemma 1, the follower maximizes $(a - q_F)\, q_F - C_F(q_F) - K_F$.

Lemma 2 (Follower's best response)**.** *Under condition* (2)*, the follower's problem has a unique solution for every* $q_L \geq 0$*:*

$$q_F^*(a) = \begin{cases} 0 & \text{if } a \leq c_F, \\ \dfrac{a - c_F}{2} & \text{if } c_F < a \leq c_F + 2/c_F, \\ \dfrac{a + \sqrt{a^2 - 8}}{4} & \text{if } a > c_F + 2/c_F. \end{cases} \tag{6}$$

The follower invests if and only if $a > c_F + 2/c_F$*. The best response* q_F^* *is continuous and nonincreasing in* q_L*. In the region in which the follower invests,*

$$\frac{dq_F^*}{dq_L} = -\frac{1}{4}\left(1 + \frac{a}{\sqrt{a^2 - 8}}\right) \in \left(-1,\ -\tfrac{1}{2}\right). \tag{7}$$

The market price $p(q_L) \equiv d - q_L - q_F^*(d - q_L)$ *is continuous and strictly decreasing in* q_L*.*

Proof. See Appendix A. □

Without investment, the follower reduces its output by one half unit for every additional unit of leader output, as in the standard Stackelberg model. With investment, the reduction lies between one half and one unit: when the leader expands, the follower's variable cost base shrinks, the follower invests less, and its output falls further.

Leader. Anticipating the follower's best response and choosing investment according to Lemma 1, the leader solves

$$\max_{q_L \in [0,d]} \ \Pi_L(q_L) \equiv p(q_L)\, q_L - C_L(q_L) - K_L. \tag{8}$$

Because the follower's investment response makes the leader's residual demand convex in some regions, Π_L need not be concave. When demand is low, for example, the leader may find it profitable to expand its output so much that the follower stops investing. We rule out such outcomes

and assume that demand is sufficiently large relative to unit variable costs. Conditions (13) in Appendix A state this formally.

Proposition 1 (Equilibrium)**.** *Under conditions* (2) *and* (13)*, the game has a unique subgame-perfect equilibrium. In this equilibrium:*

(a) Both firms produce, and the follower invests, $I_F^* = \ln(c_F q_F^*) > 0$.

(b) The leader invests if and only if $c_L q_L^* > 1$*, in which case* $I_L^* = \ln(c_L q_L^*)$.

(c) For a given demand level d*, equilibrium outputs and the market price are the same for all cost parameters* (c_L, c_F, K_L, K_F) *that satisfy conditions* (2) *and* (13) *and for which the leader invests in equilibrium.*

Proof. See Appendix A. □

Proposition 1 has an important implication for what follows. As long as the leader invests, differences in cost structure do not affect equilibrium outputs. They are absorbed entirely by investment, $I_i^* = \ln c_i + \ln q_i^*$, so that, at a given scale, the firm with the lower unit variable cost invests less.

2.4 Demand Shocks and Investment

We now consider an unanticipated decline in demand from d_0 to $d_1 < d_0$. We refer to equilibrium investment at d_0 as *planned investment*, I_i^0, and to equilibrium investment at d_1 as *revised investment*, I_i^1. For a firm with positive planned investment, its *revision share*, i.e., the share of planned investment that it postpones or cancels, is

$$\sigma_i \equiv \frac{I_i^0 - I_i^1}{I_i^0}. \tag{9}$$

Throughout, we consider demand shocks after which demand remains sufficiently large, in the sense that conditions (13) hold at every demand level $d \in [d_1, d_0]$.

Because the follower's response to the leader is nonlinear, it is not immediate that both firms produce less when demand falls. Lemma 3 establishes this, together with the corresponding decline in profits.

Lemma 3 (Demand, outputs, and profits)**.** *Suppose conditions* (2) *and* (13) *hold for all* $d \in [d_1, d_0]$*. Then the equilibrium outputs* $q_L^*(d)$ *and* $q_F^*(d)$ *and both firms' equilibrium profits are strictly increasing in* d *on* $[d_1, d_0]$*. Moreover,* $q_L^*(d_0) - q_L^*(d_1) < d_0 - d_1$.

Proof. See Appendix A. □

We write q_i^t for firm i's equilibrium output at demand d_t, $t \in \{0,1\}$, and $x_i \equiv \ln(q_i^0/q_i^1) > 0$ for the proportional decline in its output.

Proposition 2 (Investment revisions of the leader and the follower)**.** *Suppose conditions* (2) *and* (13) *hold for all* $d \in [d_1, d_0]$ *and the leader invests at* d_0*.*

(a) Both firms reduce their investment. The follower invests at both demand levels, and its revision share is $\sigma_F = x_F / \ln(c_F\, q_F^0) \in (0,1)$*.*

(b) If the leader also invests at d_1*, its revision share is* $\sigma_L = x_L / \ln(c_L\, q_L^0)$*, and*

$$\sigma_L > \sigma_F \quad \Longleftrightarrow \quad c_L < \bar{c} \equiv \frac{\left(c_F\, q_F^0\right)^\theta}{q_L^0}, \qquad \theta \equiv \frac{x_L}{x_F}. \tag{10}$$

The outputs q_L^t *and* q_F^t*, and hence* θ *and* $\bar{c}$*, are the same for all cost parameters that satisfy conditions* (2) *and* (13) *and for which the leader invests at* d_0 *and* d_1*. On this set,* σ_L *is strictly decreasing in* c_L*.*

(c) Suppose that, in a market that differs only in the leader's cost structure, a leader with a unit variable cost $c_L' \in [c_L, c_F)$ *satisfies conditions* (2) *and* (13) *for all* $d \in [d_1, d_0]$ *and invests at both demand levels, and let* $\hat{q}_L^1$ *denote its output at* d_1*. Then the leader stops investing at* d_1 *if and only if* $c_L \leq 1/\hat{q}_L^1$*, in which case* $\sigma_L = 1 > \sigma_F$*. The threshold* $1/\hat{q}_L^1$ *lies below the threshold* $\bar{c}$ *of that market.*

Proof. See Appendix A. □

Proposition 2 is the model's main result. A negative demand shock reduces both firms' output and thereby the cost base to which cost reductions apply, so both firms scale back their investment programs. As long as a firm continues to invest, its investment falls by the proportional decline in its output, x_i, irrespective of its cost structure (Lemma 1). What differs is the size of the program from which this reduction is taken. Because the leader already produces at a low unit variable cost, its program is comparatively small. If its cost advantage is large enough, a reduction of similar size therefore represents a larger share of its planned investment than of the follower's (part (b)). With an even larger cost advantage, the leader's program is so small that the shock eliminates it: the leader stops investing, whereas the follower continues to invest (part (c)).

Table 1 illustrates Proposition 2 for a decline in demand from $d_0 = 50$ to $d_1 = 38$ when the follower's unit variable cost is $c_F = 0.7$, so that $\bar{c} = 0.33$. The follower postpones or cancels 12.8

Table 1: Numerical Illustration of Proposition 2

Firm	Unit variable cost c_i	Output at d_0	Output at d_1	Planned investment I_i^0	Revised investment I_i^1	Share revised σ_i
Follower	0.70	12.44	9.42	2.164	1.886	12.8%
Leader	0.30	25.04	19.05	2.017	1.743	13.5%
Leader	0.10	25.04	19.05	0.918	0.645	29.8%
Leader	0.05	25.04	19.06	0.225	0.000	100.0%

Notes: Demand falls from $d_0 = 50$ to $d_1 = 38$, and the follower's unit variable cost is $c_F = 0.7$. In this market, $\theta = 0.98$, $\bar{c} = 0.33$, and $1/\hat{q}_L^1 = 0.052$. The follower's figures are the same for every leader that invests at both demand levels (Proposition 1(c)). The leader with $c_L = 0.05$ stops investing at d_1, so its output at d_1 differs slightly from that of the investing leaders, and the follower's revision share in that market differs by less than 0.01 percentage points. Committed fixed costs K_i do not affect any of the reported quantities as they are sunk.

percent of its planned investment. Leaders with $c_L = 0.3$ and $c_L = 0.1$ revise 13.5 and 29.8 percent, respectively, and a leader with $c_L = 0.05$ stops investing altogether.

2.5 Predictions

Economically, the result reflects that past commitments and current investment are substitutes. A firm with a rigid cost structure has already realized much of the cost savings available to it, so a larger part of its remaining investment program consists of projects that are only marginally profitable at the planned volume. When demand falls, cost reductions apply to fewer units, and marginally profitable projects are the first to be postponed or canceled. Because such projects account for a larger part of the rigid firm's program, it cuts a larger share of its planned investment. For a firm with a more flexible cost structure, the same reduction affects only a small part of its program. This mechanism does not rely on financing frictions. In our model, firms face no financing constraints and committed costs do not enter any decision as they are already sunk. We summarize the model's predictions in two hypotheses. The first concerns the direction of the effect:

Hypothesis 1. After a negative demand shock, firms with higher operating leverage reduce their investment by more than firms with lower operating leverage: they postpone or cancel a larger share of their planned investment and are less likely to continue investing at all.

The model predicts this effect when the difference in cost structure is sufficiently large ($c_L < \bar{c}$ in Proposition 2). Within this range, the share of planned investment that the leader revises increases steadily as its cost structure becomes more rigid. The second hypothesis captures this gradient:

Hypothesis 2. After a negative demand shock, the share of planned investment that firms postpone or cancel increases with their degree of operating leverage.

3 Data and Research Design

3.1 Data

Testing how operating leverage shapes investment adjustment in a crisis requires three pieces of information for the same firm: the magnitude of the demand shock it experienced, the extent to which that shock passed through to its profit, and the revision of the investment program it had planned before the crisis. The large-scale surveys conducted by the German Business Panel (GBP) provide all three. The GBP is a survey institute that samples from the universe of legal entities in the German Statistical Business Register. Since 2020, it has surveyed key decision-makers of German firms in recurring waves on firm policies relating to accounting and taxation, economic expectations, and business outcomes. Bischof et al. (2025) show that respondents closely match the register population in their distribution across industries, size classes, and legal forms.

Our analysis uses the first two GBP survey waves, both fielded after the outbreak of the COVID-19 pandemic. Germany imposed its first nationwide shutdown on 23 March 2020 and began reopening stores from 20 April, with schools following over the subsequent weeks. The first wave ran from July to October 2020, between this reopening and the renewed restrictions of November. The second was in the field from November 2020 to June 2021 and thus spans the partial shutdown of 2 November, the second lockdown of 16 December, and its extension into the following spring. Both survey waves therefore reached firms while their response to the shock was still forming rather than in retrospect. This timing is what makes our dependent variables observable. By contrast, archival data cannot support this analysis, because they record only realized investment and not the plans that firms formed before the crisis and subsequently postponed or canceled.

Our final sample comprises 6,786 firms from the GBP's first wave and 8,372 from the second. Table A1 documents each step of the sample selection. Of the 19,831 firms responding to the first wave, we retain those reporting their revenue and profit changes and answering the items on crisis responses, government support, and plans for investment and hiring. We additionally require a non-zero revenue change because our measure of operating leverage is not defined without it. Of the 6,786 firms retained, 4,873 report a revenue decline and 1,913 an increase. The second wave enters our design primarily through the validation of the operating leverage measure and therefore requires only a non-zero revenue change and a profit change. Of these firms, 5,885 report a decline. The two waves overlap in 1,634 firms that report a revenue decline in the first wave and participate again in the second. These panel firms allow us to validate our measure of operating leverage across survey waves.

The composition of the sample mirrors the population of German firms, in which privately held small and medium-sized businesses predominate. Table A2 reports the sample by industry, size, and legal form, together with the exposure of each group to the crisis. 74 percent of the firms in our sample report annual revenue below two million euros, 63 percent employ fewer than ten people, and 71 percent are corporations. Firms of this size file abridged accounts that disclose neither the composition of their costs nor their investment plans, so the adjustment we study is largely unobserved in public data. Across these firms, exposure to the crisis varied with a firm's line of business rather than with its size. Among the larger industry sections, the share of firms reporting a revenue decline ranges from 91 percent in arts and entertainment, 88 percent in accommodation and food services, and 84 percent in transportation down to 54 percent in construction. Across size classes it is almost flat, between 68.9 and 76.7 percent in every employee and revenue class.

One concern specific to survey data is that responses are uninformative if the respondent does not hold a position from which the firm's key performance indicators and its strategic responses can be reported reliably.[3] Table A3 alleviates this concern, describing who completed the questionnaire for the 4,873 firms with a revenue decline. The reported figures originate predominantly from the firms' key decision-makers: among firms that state a position, 93 percent were answered by an owner, managing director, board member or authorized signatory. Neither the precision of the answers nor the resulting classification into high or low operating leverage varies with that position. The share of firms whose revenue change is a round multiple of ten, a measure of how precisely the figure is given, lies between 44 and 54 percent in every group, and the share classified as having high operating leverage between 39 and 50 percent. Tested jointly, the positions do not predict how a firm is classified ($F = 0.85$, $p = 0.49$). Firms that leave the position blank, 24 percent of the sample, are almost identical to the owner-answered firms in both respects.

3.2 Research Design

Operating leverage cannot be computed from the financial statements available for our sample, so we construct a firm-level measure from the survey's quantitative items. Section 3.2.1 develops this measure and specifies the regression relating operating leverage to investment revisions. Section 3.2.2 compares firms across the two resulting groups, and Section 3.2.3 validates the classification against firms' own assessments of their cost structure collected in the second survey wave.

[3] If responses are uninformative in the sense that they carry noise, they would bias our estimates toward zero. This would differ if a reporting error were systematically linked to a respondent's position, for instance if respondents in a certain position were systematically more negative.

3.2.1 Identification

Operating leverage is the sensitivity of a firm's profit to its revenue and it arises from the composition of the firm's costs: the larger the share of costs that does not fall when output falls, the more of a revenue decline passes through into profit. Archival studies measure it either from the composition of a firm's costs (Lev, 1974; Novy-Marx, 2011; Aboody et al., 2018) or from the elasticity of operating income to sales, estimated over a time series of annual observations (Mandelker and Rhee, 1984; O'Brien and Vanderheiden, 1987). Both routes require detailed cost information or repeated financial statements, which the firms in our sample generally do not publish (Kallapur and Eldenburg, 2005; Minnis and Shroff, 2017).

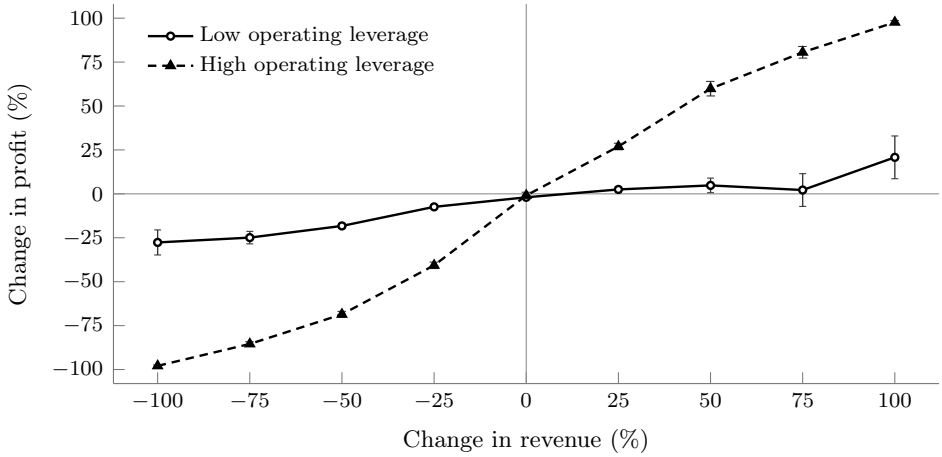
We therefore rely on survey data rather than on public financial statements. The survey does not elicit a decomposition of costs into fixed and variable components, which firm decision-makers could not readily report. Instead, it records the change in revenue and the change in profit for every firm in our sample. From these we infer operating leverage through the property that defines it, the sensitivity of profit to revenue. The ratio of the two alone would not isolate the construct, however, because the pass-through of a revenue decline to profit also varies with firm characteristics. For the 6,786 firms in our first-wave sample, we estimate a benchmark regression that predicts, for each firm i, the profit change expected given its reported revenue change, its industry membership and its size class:

$$\Delta\text{Profit}_i \;=\; \lambda_0 \;+\; \lambda_1\,\Delta\text{Revenue}_i \;+\; \mathbf{z}_i'\boldsymbol{\lambda}_2 \;+\; \big(\mathbf{z}_i\,\Delta\text{Revenue}_i\big)'\boldsymbol{\lambda}_3 \;+\; \mu_{m(i)} \;+\; u_i, \qquad (11)$$

where $\mathbf{z}_i$ is a vector of indicators for the firm's two-digit industry and for its revenue and employee size classes, $\mathbf{z}_i\,\Delta\text{Revenue}_i$ interacts each of them with the revenue change, and $\mu_{m(i)}$ is a fixed effect for the month in which the firm answered. The regression delivers a firm-specific benchmark: each firm is assessed against the profit change typical of firms in its industry and size class that report the same revenue change. A firm whose revenue did not change offers no pass-through to observe, which is why we exclude those firms.

Among firms that experienced a revenue decline, we classify a firm as having *high operating leverage* if its residual $\hat{u}_i$ from (11) is negative, that is, if its profit fell by more than predicted given its revenue change and observable characteristics, and as having *low operating leverage* if its profit fell by less. Conversely, among firms whose revenue rose, high operating leverage means that profit rose by more than predicted, while low operating leverage means that it rose by less. The resulting measure is relative rather than absolute: by construction, close to half of the firms with a revenue

Figure 1: Construction of the Operating Leverage Measure



This figure reports the average change in profit within bins of the revenue change, separately for firms with high and with low operating leverage. A firm is classified as having high operating leverage if its profit moved further than predicted by the first-stage regression in Section 3.2.1: downwards by more than predicted after a revenue decline, upwards by more than predicted after a revenue increase. The bins are 25 percentage points wide and centered on the marked values, and cover all 6,786 firms of the first wave, 3,319 of them with low and 3,467 with high operating leverage. The bars are 95 percent confidence intervals for the group mean.

decline fall on either side of the classification. It therefore ranks firms within the sample rather than locating them on an absolute scale of operating leverage.

Figure 1 illustrates the mechanics of the classification by plotting the average change in profit against the change in revenue for the two operating-leverage groups across the full first survey wave. Profit responds more strongly to revenue changes among firms classified as having high operating leverage. A 50 percent revenue decline, for example, is associated with a profit decline of about 69 percent for high-operating-leverage firms, compared with about 18 percent for low-operating-leverage firms, while the pattern reverses when revenue increases. The measure therefore captures the central economic feature of operating leverage: a cost structure that amplifies the effect of revenue changes on profit, generating larger profit declines when revenue contracts and larger profit gains when revenue expands.

The classification is the input to the second step of our design. We relate it to firms' revisions of their investment plans using

$$y_i \;=\; \beta_0 \;+\; \beta_1\,\text{High operating leverage}_i \;+\; \mathbf{x}_i'\boldsymbol{\beta}_2 \;+\; e_i, \tag{12}$$

estimated by OLS on the 4,873 firms that experienced a revenue decline.[4] The primary outcomes y_i are the shares of planned investment that the firm postpones and cancels, and indicators for

[4]In our main analysis, we focus on the firms with declining revenues during the pandemic. In Section 4.5, we also consider firms that experienced a positive revenue shock during COVID-19.

whether the firm plans to invest over the next twelve months and over the subsequent twelve to twenty-four months. The controls $\mathbf{x}_i$ comprise the revenue change, entering as a separate fixed effect for each reported value; fixed effects for the revenue and employee size classes; the firm's expected GDP growth for 2020 and 2021; and industry fixed effects. We report heteroskedasticity-robust standard errors throughout. Because firms in the same industry may be exposed to correlated shocks, Section 4.2 repeats the estimation with standard errors clustered at the two-digit industry level, alongside alternative first stages, control sets, samples, and estimators.

The design compares firms exposed to similar revenue declines but whose profits responded differently to those declines. The first-stage residual identifies this differential profit response, while conditioning on the revenue change in (12) ensures that β_1 does not simply reflect differences in the severity of the revenue shock across the two operating-leverage groups. The remaining controls account for differences in industry, firm size, and expectations about the recovery. This specification maps directly onto the model in Section 2. Hypothesis 1 predicts that firms with higher operating leverage revise a larger share of their investment program: $\beta_1 > 0$ for postponed and canceled investment and $\beta_1 < 0$ for short- and medium-term investment. Hypothesis 2 concerns the gradient in this relation rather than its sign. Testing it therefore requires a measure of degree rather than the binary classification used in (12). Section 4.1 divides firms according to the distance between their realized and predicted profit changes and tests whether the association with investment revisions strengthens with this distance.

3.2.2 Descriptive Statistics

Table 2 reports descriptive statistics for the 4,873 firms with a revenue decline, separately for firms classified as having high and low operating leverage.[5] Panel A reports pre-crisis firm characteristics. Firms with high operating leverage are somewhat smaller than firms with low operating leverage, with mean revenue of 3.3 versus 4.5 million euros and 29 versus 39 employees, and are slightly more concentrated in services and less in manufacturing. Although several of these differences are statistically significant, their magnitudes are economically limited: normalized differences do not exceed 0.12 in absolute value, well below the threshold of 0.25 at which differences in observable characteristics may become large enough to make estimates sensitive to how those characteristics are controlled for (Imbens and Wooldridge, 2009).

Panel B describes firms' exposure to the crisis and their expectations at the time of the survey. Firms classified as having high operating leverage report a larger revenue decline on average, which

[5]Table A4 records detailed definitions for all variables used in the analyses.

Table 2: Descriptive Statistics – Negative Revenue Shock

	Full sample			Operating leverage		Difference (high – low)		
	Mean	Median	Std.	Low	High	Mean diff.	t-statistic	Norm. diff.
Panel A: Pre-crisis firm characteristics								
Revenue 2019 (EUR m)	3.90	0.52	10.25	4.50	3.28	−1.21	-4.12^{***}	−0.12
No. of employees	33.72	3.00	108.10	38.55	28.69	−9.85	-3.16^{***}	−0.09
Manufacturing	0.15	–	–	0.16	0.13	−0.02	-2.18^{**}	−0.06
Trade	0.15	–	–	0.16	0.15	−0.02	-1.69^{*}	−0.05
Services	0.64	–	–	0.61	0.66	0.04	3.17^{***}	0.09
Panel B: Crisis exposure and expectations								
Δ Revenue (%)	−43.64	−40.00	28.75	−36.59	−50.97	−14.38	-18.03^{***}	−0.52
Δ Profit (%)	−37.83	−28.00	37.50	−12.27	−64.36	−52.09	-67.37^{***}	−1.92
Δ Employees (%)	−4.66	0.00	29.70	−2.01	−7.41	−5.40	-6.37^{***}	−0.18
Δ Liquidity (%)	−24.42	−20.00	34.98	−13.74	−35.49	−21.74	-22.72^{***}	−0.65
Δ Liabilities (%)	3.96	0.00	28.99	3.22	4.72	1.50	1.80^{*}	0.05
Government aid received	0.76	–	–	0.72	0.80	0.08	6.84^{***}	0.20
Expected GDP growth 2020 (%)	−6.84	−7.00	2.93	−6.50	−7.19	−0.68	-7.75^{***}	−0.24
Expected GDP growth 2021 (%)	−0.83	0.00	4.86	−0.90	−0.76	0.14	0.94	0.03
Panel C: Outcomes								
Pre-crisis investment postponed (share)	0.28	0.10	0.34	0.25	0.32	0.07	7.30^{***}	0.21
Pre-crisis investment canceled (share)	0.19	0.00	0.30	0.14	0.24	0.10	11.28^{***}	0.32
Investment plans, 0–12 months	0.37	–	–	0.41	0.34	−0.07	-5.28^{***}	−0.15
Investment plans, 12–24 months	0.53	–	–	0.58	0.49	−0.09	-6.24^{***}	−0.18
Hiring plans, 0–12 months	0.25	–	–	0.29	0.20	−0.09	-7.29^{***}	−0.21
Hiring plans, 12–24 months	0.41	–	–	0.44	0.38	−0.07	-4.89^{***}	−0.14
Layoffs	0.23	–	–	0.19	0.27	0.08	6.48^{***}	0.19
Price increases	0.24	–	–	0.23	0.25	0.02	1.74^{*}	0.05
N	4,873			2,482	2,391			

This table presents descriptive statistics for firms that experienced a negative revenue shock following the onset of the COVID-19 pandemic. Firms are classified as having high (low) operating leverage if their profit declined by more (less) than predicted by the first-stage regression in Section 3.2.1. Panel A reports firm characteristics before the crisis, Panel B the crisis exposure and expectations at the time of the survey, and Panel C the outcome variables. Changes are measured relative to January 2020. Revenue in 2019 and the number of employees are reported in size classes, so their statistics use class midpoints, with the top class set to its lower bound. The normalized difference is the mean difference divided by $\sqrt{(s^2_{high} + s^2_{low})/2}$. The difference in the profit change between the groups follows from the construction of the classification. Services do not include wholesale and retail trade, which is reported separately; the remaining firms operate in agriculture, mining, energy and water supply, or construction. The sector indicators are available for 4,704 firms with known industry classification, the expected GDP growth for 2020 (2021) for 4,343 (4,309) firms. The stars refer to t-tests of the difference in means. Please refer to Appendix A4 for a full description of all variables. ***, **, * indicate statistical significance at the 1%, 5%, and 10% levels (two-tailed), respectively.

is why (12) conditions on the magnitude of the revenue change. The difference in profit declines follows by construction, as illustrated in Figure 1. The groups also differ in how the crisis affected employment and liquidity. Firms with high operating leverage report declines in employment of 7.4 versus 2.0 percent and in liquidity of 35.5 versus 13.7 percent. They are also more likely to have claimed a government support measure, at 80 versus 72 percent. The difference in liquidity is consistent with operating leverage placing greater financial pressure on firms when revenue contracts, but also raises the possibility that differences in financial constraints contribute to subsequent investment decisions. Section 4.3 examines this possibility directly.

Panel C reports the outcome variables and provides first univariate evidence of differences between the two groups. Firms with high operating leverage scale back their investment plans more substantially. They postpone 32 versus 25 percent of planned investment and cancel 24 versus 14 percent, and are 7 and 9 percentage points less likely to plan investment over the next twelve and the subsequent twelve to twenty-four months, respectively. All four differences are significant at the one percent level. This pattern extends to other firm decisions: firms with high operating leverage are less likely to hire, more likely to lay off employees, and slightly more likely to increase prices. Section 4 examines whether these differences persist after accounting for the controls in (12).

3.2.3 Validation

Our measure of operating leverage is inferred from firms' reported changes in revenue and profit rather than directly from their cost structures. We therefore examine whether the classification captures the intended construct. One concern is that the residual profit response used to classify firms could reflect factors other than cost rigidity that are themselves related to subsequent investment decisions. For example, firms with more pessimistic expectations about the recovery could report larger profit declines and simultaneously reduce investment, generating the predicted relation even if the classification did not capture differences in operating leverage.[6] The GBP's second survey wave allows us to validate the measure against firms' reported cost flexibility. Firms were asked how difficult it would be to reduce costs within three months following a decline in demand, on a scale from zero to ten, and which obstacles prevent them from doing so. If the classification captures operating leverage, firms classified as high operating leverage in the first wave should report greater difficulty reducing costs.

[6] In addition to the validation tests below, our main regressions control for firms' expectations about GDP growth over the following two years.

Table 3: Validation – Difficulty of and Hurdles to Cost Reduction

		Hurdles to cost reduction						
	Difficulty of cost reduction (0–10)	Long-term commitments	Asset disposal	Planning uncertainty	Legal obligations	Know-how retention	Employee contracts	No hurdles
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
High operating leverage (wave 1)	0.534***	0.058**	0.043**	0.030	0.029	0.021	−0.005	−0.023*
	(0.135)	(0.026)	(0.019)	(0.026)	(0.023)	(0.024)	(0.023)	(0.012)
Mean of low group	6.486	0.509	0.154	0.484	0.267	0.534	0.273	0.068
R^2	0.132	0.097	0.124	0.100	0.143	0.237	0.160	0.114
N	1,618	1,610	1,610	1,610	1,610	1,610	1,610	1,610
Revenue change (wave 1) × revenue size class	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Size classes, GDP expectations, industry FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

This table relates the classification of operating leverage to the firms' own assessment of their cost flexibility. Firms are classified in the first survey wave and give the assessment in the second (November 2020 to June 2021), so that the two are separated by several months and by the questionnaire in which they were collected. The sample comprises the 1,634 panel firms that participated in both waves and reported a revenue decline in the first, irrespective of their revenue change in the second. In column (1), the dependent variable is the stated difficulty of reducing costs within three months when demand decreases (0 = very easy, 10 = very difficult). In columns (2) to (8), the dependent variables are indicators for the hurdles to cost reduction that firms indicate, with multiple answers possible. The hurdles are ordered by the size of the difference between firms with high and low operating leverage. The regressions control for the first-wave revenue change, linearly and interacted with the revenue size class, and for the revenue and employee size classes, the expected GDP growth, and industry fixed effects. Table A5 reports the same comparison with firms classified and assessed within the second wave only. Please refer to Appendix A4 for a full description of all variables. Heteroskedasticity-robust standard errors are reported in parentheses. ***, **, * indicate statistical significance at the 1%, 5%, and 10% levels (two-tailed), respectively.

Table 3 reports the results of this validation test for the firms that participated in both the first and second survey waves.[7] The panel structure allows us to validate the measure using information collected separately from that used to construct it. Specifically, we retain each firm's operating leverage classification from the first wave and relate it to the difficulty of reducing costs reported in the subsequent wave. This separation reduces the concern that an association between the two measures arises mechanically from contemporaneous reporting behavior or wave-specific measurement error. Consistent with the classification capturing differences in cost rigidity, firms classified as having high operating leverage rate the difficulty of reducing costs 0.53 points higher on the eleven-point scale, a difference significant at the one percent level.[8] Relative to the average rating of 6.5 among firms with low operating leverage, this difference implies an increase of approximately 8 percent relative to the group's mean.

The individual obstacles to cost reduction provide further evidence on what the measure captures. The two groups differ primarily in long-term commitments and the uneconomical disposal of assets, with both differences significant at the five percent level. These obstacles reflect commitments made before the crisis and are therefore consistent with the measure capturing predetermined differences in cost rigidity rather than merely differences in firms' expectations about the recovery. Employee contracts, in contrast, do not distinguish the two groups (column (7)). This absence is consistent with the institutional setting: short-time work allowances were widely available during this period, allowing German firms to temporarily adjust labor costs that would otherwise have been more difficult to unwind. Finally, consistent with the overall difference in the difficulty of reducing costs, firms with high operating leverage are less likely to report facing no obstacles to cost reduction (column (8)). Taken together, the results indicate that our measure captures underlying differences in cost structures associated with commitments made before the crisis.

4 Results

4.1 Operating Leverage and Investment Revisions

Table 4 tests our two hypotheses using the 4,873 firms that experienced a revenue decline. Hypothesis 1 predicts that firms with higher operating leverage scale back a larger share of their

[7] Table A5 repeats the analysis within the second survey wave, applying (11) to firms' second-wave responses. The results are comparable to those for the panel firms.

[8] The result does not depend on the binary classification of operating leverage. Replacing the classification with the standardized negative residual, the continuous measure used in Table 4, Panel B, shows that a one-standard-deviation increase in operating leverage is associated with a 0.24-point increase in the reported difficulty of reducing costs ($p < 0.01$).

Table 4: Investment Revisions – Negative Revenue Shock

	Pre-crisis investment plans		Future investment plans	
	Postponed (share) (1)	Canceled (share) (2)	0–12 months (indicator) (3)	12–24 months (indicator) (4)
Panel A: High versus low operating leverage				
High operating leverage	0.048***	0.059***	−0.036***	−0.046***
	(0.010)	(0.009)	(0.014)	(0.014)
Mean of low group	0.246	0.143	0.411	0.576
Effect relative to mean (%)	19.6	41.3	−8.9	−8.0
R^2	0.098	0.165	0.113	0.134
Panel B: Degree of operating leverage				
Operating leverage (standardized)	0.029***	0.030***	−0.008	−0.009
	(0.005)	(0.004)	(0.007)	(0.007)
Mean of all firms	0.280	0.190	0.375	0.532
Effect relative to mean (%)	10.2	15.9	−2.2	−1.8
R^2	0.101	0.166	0.112	0.132
Revenue change fixed effects	Yes	Yes	Yes	Yes
Revenue size class	Yes	Yes	Yes	Yes
Employee size class	Yes	Yes	Yes	Yes
Expected GDP growth 2020/2021	Yes	Yes	Yes	Yes
Industry fixed effects	Yes	Yes	Yes	Yes
N	4,873	4,873	4,873	4,873

This table reports OLS estimates of investment revisions and investment plans on operating leverage for firms that experienced a negative revenue shock. Firms are classified as having high (low) operating leverage if their profit declined by more (less) than predicted by the first-stage regression in Section 3.2.1. In Panel A, operating leverage enters as the indicator for the high group. In Panel B, it enters as the negative of the first-stage residual, standardized to zero mean and unit variance over the estimation sample, so that the coefficient is the effect of a one standard deviation increase in the degree of operating leverage. In columns (1) and (2), the dependent variable is the share of the investment planned before the crisis that the firm postpones or cancels; in columns (3) and (4), it is an indicator for current plans to invest within the next 12 months or in 12 to 24 months. All regressions control for the revenue change with a separate fixed effect for each reported value, and for the revenue and employee size classes, the expected GDP growth for 2020 and 2021, and industry fixed effects. The effect relative to the mean is the coefficient divided by the mean reported in the same panel. Table A6 reports Panel B in terciles of the degree. Please refer to Table A4 for a full description of all variables. Heteroskedasticity-robust standard errors are reported in parentheses. ***, **, * indicate statistical significance at the 1%, 5%, and 10% levels (two-tailed), respectively.

planned investment and are less likely to continue investing, while Hypothesis 2 predicts that these responses become stronger with the degree of operating leverage. Columns (1) and (2) examine the shares of the pre-crisis investment program that firms postpone and cancel, and columns (3) and (4) examine whether firms plan to invest over the next twelve and the subsequent twelve to twenty-four months.

Panel A begins with the binary classification of operating leverage, estimating (12). The results support Hypothesis 1 across all four outcomes. Firms with high operating leverage postpone 4.8 and cancel 5.9 percentage points more of their planned investment than firms with low operating leverage, and are 3.6 and 4.6 percentage points less likely to plan investment over the two horizons. Relative to the means of the low operating leverage group, these differences correspond to 19.6 and

41.3 percent more investment postponed and canceled, respectively, and 8.9 and 8.0 percent lower probabilities of planning investment. All four estimates are significant at the one percent level.[9] Importantly, the specification includes separate fixed effects for each reported revenue change. Thus, conditional on the remaining controls, β_1 is identified from comparisons between firms reporting the same revenue change but differing in their operating leverage classification.

Panel B turns to Hypothesis 2 and replaces the binary classification with the distance between the firm's realized and predicted profit change, standardized to zero mean and unit variance. A one-standard-deviation increase in operating leverage raises the share of planned investment postponed by 2.9 percentage points and the share canceled by 3.0 percentage points, corresponding to 10.2 and 15.9 percent of their respective sample means. Both estimates are significant at the one percent level. Thus, the extent to which firms revise their planned investment increases with the degree of operating leverage.[10] The two investment-plan indicators, however, do not vary systematically with the degree of operating leverage. Thus, we find the gradient predicted by Hypothesis 2 in how much planned investment firms postpone or cancel, but not in whether they plan to invest over the following two years.

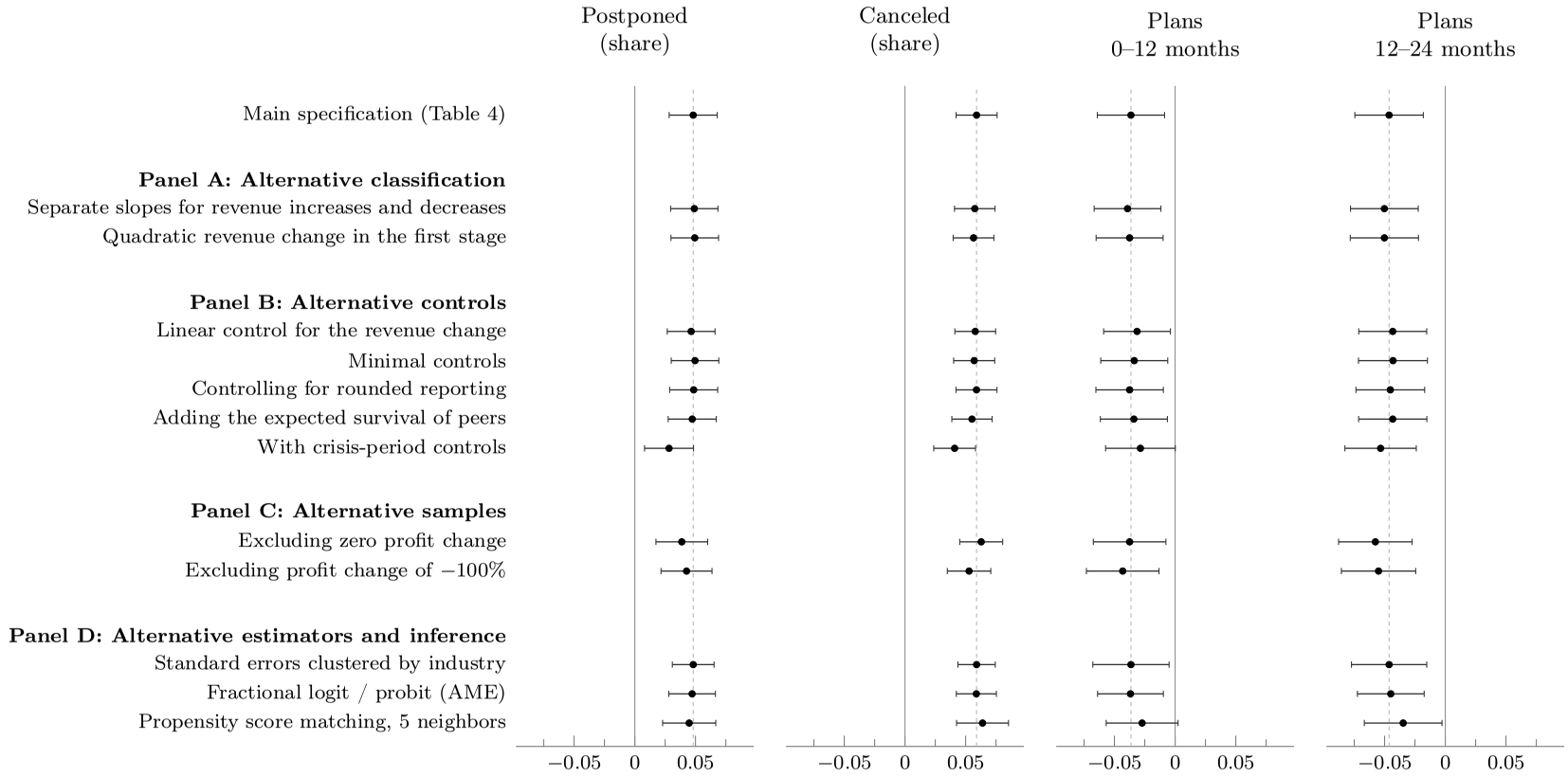
4.2 Robustness

Varying the Specification. Figure 2 reports the coefficient on high operating leverage across multiple variations of the specification in Table 4, with Appendix Table A7 reporting the estimates, standard errors, and sample sizes. Panel A examines whether the results depend on the specification used to classify firms' operating leverage. Because our measure is based on the residual from (11), misspecification of the relation between revenue and profit changes could affect the classification. For example, if this relation differs between firms with revenue increases and decreases, imposing a common slope could systematically over- or underpredict profit changes and thereby affect the residual. We therefore reclassify firms twice, first allowing the intercept and slope to differ with the direction of the revenue change and then adding a quadratic term. Neither reclassification changes any of the four main coefficients by more than 0.4 percentage points.

[9] The association holds in both halves of the size distribution. Among the 4,804 firms that report an employee size class, splitting at ten employees, firms with high operating leverage postpone 6.5 percentage points more of their planned investment among the 1,671 larger firms and 4.0 percentage points more among the 3,133 smaller ones, and cancel 5.4 and 6.2 percentage points more respectively. No difference between the two size groups is statistically significant for any of the four outcomes.

[10] Table A6 shows that this result does not depend on imposing a linear gradient. Dividing firms with high operating leverage into terciles based on their distance from the predicted profit change, while retaining firms with low operating leverage as the reference group, the postponed share increases from 2.6 percentage points in the lowest tercile to 5.3 and 6.8 percentage points in the two higher terciles. The corresponding estimates for cancellations are 2.9, 6.6, and 8.5 percentage points. The lowest and highest terciles differ significantly for both outcomes ($p = 0.019$ and $p = 0.001$, respectively).

Figure 2: Robustness of the Association Between Operating Leverage and Investment



This figure reports the coefficient on the indicator for high operating leverage from variations of the main specification in Table 4. Each marker comes from a separate regression and the horizontal bars are 95 percent confidence intervals. The solid vertical line marks zero and the dotted vertical line the estimate of the main specification. Appendix Table A7 reports the same estimates with their standard errors as well as sample sizes and describes each variation.

Panel B examines the sensitivity of the results to the controls and potential differences in firms' reporting and expectations. First, we replace the fixed effects for each reported revenue change with a linear control interacted with revenue size class and, separately, reduce the set of controls to revenue change, firm size, and industry. The estimates remain similar under both specifications. We next address potential differences in reporting precision. The 34 percent of firms reporting both revenue and profit changes as multiples of ten are 12 percentage points more likely to be classified as having high operating leverage. Adding an indicator for such firms leaves the estimates virtually unchanged. Moreover, we account for differences in firms' expectations about the crisis. In addition to the GDP growth expectations included in the baseline specification, we control for each firm's expectation of the share of its peers that will survive the crisis. This measure provides an alternative assessment of the firm's outlook for businesses similar to itself. The estimates again remain similar.

The final specification in Panel B controls for contemporaneous changes in employment, liquidity, and liabilities, as well as receipt of government support. This is a demanding specification because these variables are themselves affected by the crisis and may partly reflect firms' responses to the revenue shock. Conditioning on them reduces the coefficient on the postponed share from 4.8 to 2.8 percentage points and that on the canceled share from 5.9 to 4.1 percentage points, but both remain significant at the one percent level. We examine the role of liquidity more directly in Section 4.3.

Panel C examines whether the results are driven by observations for which the reported profit change may be less informative about firms' underlying profit response. We first exclude all firms reporting no change in profit. Removing these observations lowers the coefficient on the postponed share from 4.8 to 3.9 percentage points and raises that on the canceled share from 5.9 to 6.3 percentage points, leaving the overall pattern unchanged. We next exclude firms reporting a 100 percent decline in profit. These observations are at the lower bound of the reported profit change, making it impossible to observe any further decline. The resulting estimates again remain close to those in Table 4, with slightly smaller coefficients for the two revision shares and slightly larger coefficients in absolute value for the two investment-plan indicators.

Panel D varies the approach to inference and estimation. We first cluster standard errors by industry to allow for correlated errors among firms exposed to common industry-level shocks. We then account for the bounded nature of the dependent variables, estimating fractional logit models for the two revision shares and probit models for the two investment-plan indicators. The resulting average marginal effects are close to the linear estimates. Finally, propensity score matching com-

pares each firm with high operating leverage to the five firms with low operating leverage that are closest in their estimated propensity score. Figure A1 shows substantial overlap in the propensity-score distributions of the two groups. Across the variations in Figure 2, the estimates remain close to the baseline, with the largest attenuation occurring when conditioning on firms' contemporaneous crisis-period outcomes; the two revision-share coefficients remain significant at the one percent level throughout.

Varying the Measure. Table 5 examines whether our results depend on measuring operating leverage through the first-stage regression. A conventional measure of operating leverage is the ratio of the percentage change in profit to the percentage change in revenue. We do not use this ratio as our primary measure for two reasons. First, we are interested in whether a firm's profit responds more strongly to a revenue shock than would be expected for a comparable firm. The first-stage regression provides this benchmark by accounting for systematic differences in the relation between revenue and profit changes across industries and firm sizes. Including industry and size fixed effects only in the investment regression would instead account for differences in investment across these groups, without benchmarking the profit response itself.

Second, dividing by the revenue change makes the ratio particularly sensitive when the revenue change is small. Among firms whose revenue fell by less than 10 percent, the ratio ranges from -1.0 at the tenth percentile to 4.0 at the ninetieth percentile, compared with 0.0 to 1.4 among firms whose revenue fell by at least 50 percent. When revenue declines are larger, the ratio and our baseline measure are closely related: among firms whose revenue fell by at least 25 percent, the correlation between the ratio and the first-stage residual is -0.90.

We nevertheless relate investment revisions directly to the ratio. Because the ratio has large mass points at zero (566 firms) and one (732 firms), we group firms into economically meaningful bands rather than quantiles. The reference group consists of the 860 firms whose profits did not decline despite a decline in revenue. The last column of Table 5 shows that the ratio aligns closely with our baseline classification. The share classified as having high operating leverage increases from 1.2 percent in the reference group to 18.7 percent among firms whose profits fell by less than revenue, 79.4 percent among firms whose profits fell proportionately, and 95.3 and 98.5 percent in the two highest bands.

The investment results are similar using the ratio. Relative to firms whose profits did not decline, firms whose profits fell more than twice as fast as revenue postpone 8.7 and cancel 9.7 percentage points more of their planned investment, compared with reference-group means of 18.9

Table 5: An Alternative Measure of the Degree of Operating Leverage

	Pre-crisis investment plans				Future investment plans				Firms	Classified high (%)
	Postponed (share)		Canceled (share)		0–12 months		12–24 months			
	Coef.	SE	Coef.	SE	Coef.	SE	Coef.	SE		
Profit did not fall (ratio ≤ 0) (reference)	0.189		0.126		0.433		0.583		860	1.2
Profit fell less than revenue ($0 <$ ratio < 1)	0.052^{***}	(0.014)	-0.002	(0.011)	-0.031	(0.021)	-0.012	(0.021)	1,745	18.7
Profit fell in proportion (ratio $= 1$)	0.034^{*}	(0.018)	0.043^{***}	(0.015)	-0.038	(0.026)	-0.037	(0.026)	732	79.4
Profit fell up to twice as fast ($1 <$ ratio ≤ 2)	0.086^{***}	(0.015)	0.059^{***}	(0.013)	-0.062^{***}	(0.023)	-0.052^{**}	(0.023)	1,205	95.3
Profit fell more than twice as fast (ratio > 2)	0.087^{***}	(0.020)	0.097^{***}	(0.018)	0.000	(0.032)	-0.016	(0.031)	331	98.5
R^2	0.101		0.167		0.114		0.133			
N	4,873									

This table reports OLS estimates of investment revisions and investment plans on an alternative measure of the degree of operating leverage for the 4,873 firms that experienced a negative revenue shock. The measure is the ratio of the reported percentage change in profit to the reported percentage change in revenue. Unlike the classification used throughout the paper, this measure is formed without the first-stage regression. The dependent variables are the share of planned investment that the firm postponed and the share it canceled, and indicators for current plans to invest within the next 12 months or in 12 to 24 months. Firms are grouped into economic bands rather than quantiles because the ratio equals exactly zero for 566 firms whose profit did not change and exactly one for 732 firms, so that quantiles would be neither of equal size nor meaningful. The first row reports the mean of each outcome in the reference band. The last column reports the share of each band that the first-stage classification assigns to high operating leverage, against 49.1 percent of the sample as a whole. All regressions control for the revenue change with a separate fixed effect for each reported value, and for the revenue and employee size classes, the expected GDP growth for 2020 and 2021, and industry fixed effects. Please refer to Table A4 for a full description of all variables. Heteroskedasticity-robust standard errors are reported in parentheses. ***, **, * indicate statistical significance at the 1%, 5%, and 10% levels (two-tailed), respectively.

and 12.6 percent. The canceled share increases monotonically across the bands, from no significant difference among firms whose profits fell by less than revenue to 4.3, 5.9, and 9.7 percentage points. The postponed share shows a less monotonic pattern: firms whose profits fell in proportion to revenue postpone 3.4 percentage points more, compared with 5.2 percentage points among firms whose profits fell by less than revenue. This attenuation at a ratio of one may reflect greater noise at this large mass point. The two investment-plan indicators show no systematic gradient across the bands, consistent with the results in Table 4.

4.3 Financial Constraints

One alternative explanation for our results is that they may reflect financial constraints. The mechanism in our model works through the desirability of investment, as a rigid cost structure changes which projects remain worth carrying out once demand falls. Alternatively, the larger decline in profits may leave firms with fewer internal resources to finance investment, causing financially constrained firms to cut investment even when projects remain desirable. Section 4.2 provides initial evidence that liquidity is unlikely to account for the entire association, as the relation between operating leverage and investment revisions persists when controlling for firms' contemporaneous change in liquidity.

Yet, controlling for the change in liquidity may not fully capture whether a firm was actually constrained. A firm may experience a large decline in liquidity and still have reserves or credit to draw on, while another may experience only a small decline but have neither. Table 6 therefore uses firms' reliance on government support during the crisis to distinguish different degrees of financial need.[11] The first group comprises the 1,186 firms that received no government support, the second the 1,899 firms that received support but state that they would have survived without it, and the third the 1,750 firms that state that they would not. The descriptive statistics are consistent with increasing financial pressure across the three groups, with average revenue declines of 29.3, 39.0, and 58.3 percent, liquidity declines of 12.0, 17.9, and 39.8 percent, and shares of firms classified as having high operating leverage of 40.5, 45.0, and 59.4 percent.

We estimate (12) with the operating leverage classification interacted with indicators for the three groups, so that each coefficient compares firms with high and low operating leverage within the same group. If financial constraints drive the association between operating leverage and

[11] Government support in the survey covers a broad set of measures. The non-tax measures include Corona emergency aid, bridging aid, short-time work and short-time work allowances, government guarantees, and KfW lending programs, while the tax measures include, among others, deferrals of tax payments. The most frequently used measures were Corona emergency aid, short-time work and short-time work allowances, and tax-payment deferrals.

Table 6: Financial Constraints as an Alternative Explanation

	Pre-crisis investment plans		Future investment plans	
	Postponed (share) (1)	Canceled (share) (2)	0–12 months (indicator) (3)	12–24 months (indicator) (4)
Effect of high operating leverage within:				
No aid received	0.038**	0.067***	−0.059**	−0.098***
	(0.018)	(0.016)	(0.028)	(0.029)
Aid, would have survived	0.047***	0.048***	−0.032	−0.045**
	(0.016)	(0.012)	(0.023)	(0.022)
Aid, would not have survived	0.043**	0.050***	−0.018	−0.013
	(0.017)	(0.016)	(0.023)	(0.024)
p-value: equal effects across groups	0.932	0.629	0.529	0.076
Mean of low group	0.246	0.143	0.411	0.576
R^2	0.110	0.176	0.116	0.135
Revenue change fixed effects	Yes	Yes	Yes	Yes
Revenue size class	Yes	Yes	Yes	Yes
Employee size class	Yes	Yes	Yes	Yes
Expected GDP growth 2020/2021	Yes	Yes	Yes	Yes
Industry fixed effects	Yes	Yes	Yes	Yes
Financial need group indicators	Yes	Yes	Yes	Yes
N	4,835	4,835	4,835	4,835

This table reports OLS estimates of investment revisions and investment plans on an indicator for high operating leverage interacted with the firm's financial need, for firms that experienced a negative revenue shock. Firms are classified as having high (low) operating leverage if their profit declined by more (less) than predicted by the first-stage regression in Section 3.2.1. Furthermore, firms are grouped by whether they received government aid and, if so, whether they state that they would have survived the crisis without it: no aid (1,186 firms), aid but would have survived (1,899 firms), and aid and would not have survived (1,750 firms). Each coefficient is the effect of high operating leverage within the respective group. The p-value refers to a joint test of equal effects of high operating leverage across the three groups. Firms that received aid but did not answer the survival question are excluded. In columns (1) and (2), the dependent variable is the share of the investment planned before the crisis that the firm postpones or cancels; in columns (3) and (4), it is an indicator for current plans to invest within the next 12 months or in 12 to 24 months. All regressions control for the revenue change with a separate fixed effect for each reported value, and for the revenue and employee size classes, the expected GDP growth for 2020 and 2021, and industry fixed effects. Please refer to Table A4 for a full description of all variables. Heteroskedasticity-robust standard errors are reported in parentheses. ***, **, * indicate statistical significance at the 1%, 5%, and 10% levels (two-tailed), respectively.

investment revisions, the effect should be strongest where financing constraints are more likely to bind. In particular, among firms receiving government support, we would expect a stronger association for those that state they would not have survived without it than for those that state they would have survived without support.

The estimates do not show this pattern. The association is present in all three groups. Firms with high operating leverage postpone 3.8, 4.7, and 4.3 percentage points more of their planned investment across the three groups and cancel 6.7, 4.8, and 5.0 percentage points more, respectively. We do not reject equality of the coefficients across groups for either outcome. In particular, the association remains statistically significant among firms that received no government support and does not become stronger among firms that report that they would not have survived without it. The two investment-plan indicators also show no increase in the association with financial need. If

anything, the association is weaker among firms that received support, and weakest among those that state they would not have survived without it, which is where a financing constraint should bind most. Taken together, the results suggest that financial constraints do not fully account for the relation between operating leverage and investment revisions, although they do not rule out a role for financing conditions more generally.[12]

4.4 Employment and Prices

If our measure captures differences in cost rigidity, the consequences of operating leverage should extend beyond firms' investment decisions. When demand falls and existing costs do not adjust proportionately, firms may respond by reducing costs along other margins. Employment provides a particularly direct test of this idea. Firms with higher operating leverage should be more likely to reduce their workforce and less likely to expand it. Firms may also attempt to protect margins through higher prices, although this response is constrained by competition and demand conditions. Table 7 examines these outcomes using the specifications from Table 4.

Panel A reports the results for the binary classification. Firms with high operating leverage are 5.4 percentage points more likely to lay off employees and 5.2 and 3.3 percentage points less likely to plan hiring over the next twelve and the subsequent twelve to twenty-four months. Relative to the means of the low operating leverage group, these differences correspond to 28.6, 18.1, and 7.4 percent. Firms with high operating leverage are also 2.3 percentage points more likely to raise prices, although this estimate is significant only at the ten percent level. Panel B replaces the binary classification with the standardized degree of operating leverage. A one-standard-deviation increase raises the probability of layoffs by 3.5 percentage points and lowers the probability of planning to hire within the next year by 1.9 percentage points. Hiring plans over the longer horizon and price increases show no association with the degree of operating leverage.

The employment results reinforce the interpretation of our measure as capturing differences in firms' cost structures. Firms with higher operating leverage are more likely to reduce employment and less likely to expand it following the demand shock, consistent with greater pressure to adjust costs elsewhere when existing costs do not fall with output. The weaker evidence on prices may reflect more limited scope to protect margins through pricing when firms face competitive pressure.

[12] A financing-constraint explanation has no clear prediction for the pattern we observe when revenues increase. Section 4.5 shows that the association between operating leverage and investment changes tends to reverse for firms experiencing positive revenue shocks. This pattern is consistent with operating leverage magnifying firms' responses to revenue shocks in both directions and is not naturally explained by a mechanism operating only through financial constraints.

Table 7: Employment and Prices – Negative Revenue Shock

	Employment			Price increases
	Layoffs	Hiring plans 0–12 months	Hiring plans 12–24 months	
	(1)	(2)	(3)	(4)
Panel A: High versus low operating leverage				
High operating leverage	0.054^{***}	-0.052^{***}	-0.033^{**}	0.023^{*}
	(0.012)	(0.012)	(0.014)	(0.013)
Mean of low group	0.189	0.289	0.445	0.232
Effect relative to mean (%)	28.6	−18.1	−7.4	10.1
R^2	0.183	0.151	0.147	0.094
Panel B: Degree of operating leverage				
Operating leverage (standardized)	0.035^{***}	-0.019^{***}	−0.009	0.002
	(0.006)	(0.006)	(0.007)	(0.006)
Mean of all firms	0.227	0.245	0.411	0.243
Effect relative to mean (%)	15.4	−7.8	−2.3	0.8
R^2	0.186	0.149	0.147	0.093
Revenue change fixed effects	Yes	Yes	Yes	Yes
Revenue size class	Yes	Yes	Yes	Yes
Employee size class	Yes	Yes	Yes	Yes
Expected GDP growth 2020/2021	Yes	Yes	Yes	Yes
Industry fixed effects	Yes	Yes	Yes	Yes
N	4,873	4,873	4,873	4,873

This table reports OLS estimates of layoffs, hiring plans, and price increases on operating leverage for firms that experienced a negative revenue shock. Firms are classified as having high (low) operating leverage if their profit declined by more (less) than predicted by the first-stage regression in Section 3.2.1. All dependent variables are indicators. The specifications are those of Table 4: in Panel A operating leverage enters as the indicator for the high group, and in Panel B as the negative of the first-stage residual, standardized to zero mean and unit variance over the estimation sample. All regressions control for the revenue change with a separate fixed effect for each reported value, and for the revenue and employee size classes, the expected GDP growth for 2020 and 2021, and industry fixed effects. The effect relative to the mean is the coefficient divided by the mean reported in the same panel. Please refer to Table A4 for a full description of all variables. Heteroskedasticity-robust standard errors are reported in parentheses. ***, **, * indicate statistical significance at the 1%, 5%, and 10% levels (two-tailed), respectively.

4.5 Positive Revenue Shocks

Operating leverage should affect firms differently depending on the direction of the revenue shock. A rigid cost structure magnifies changes in profit in both directions, producing larger profit declines when revenue falls and larger profit increases when revenue rises because costs do not adjust proportionately with revenue. Investment responses should therefore move in opposite directions across negative and positive revenue shocks. This implication provides a useful check on what our classification captures, since interpretations based on firms' exposure to the crisis or financial constraints would not naturally predict an opposite association among firms whose revenues increased. Table 8 examines the 1,913 firms reporting a revenue increase and applies the mirror of our baseline classification, defining high operating leverage as a profit increase that exceeds the increase predicted by the first-stage regression.

Table 8: Operating Leverage and Investment – Positive Revenue Shock

	Pre-crisis investment plans		Future investment plans	
	Postponed (share) (1)	Canceled (share) (2)	0–12 months (indicator) (3)	12–24 months (indicator) (4)
High operating leverage	−0.029** (0.013)	−0.012 (0.009)	0.062** (0.024)	0.051** (0.023)
p-value: same effect after a revenue decline	0.000	0.000	0.000	0.000
Mean of low group	0.137	0.068	0.548	0.659
R^2	0.162	0.192	0.171	0.174
Revenue change fixed effects	Yes	Yes	Yes	Yes
Revenue size class	Yes	Yes	Yes	Yes
Employee size class	Yes	Yes	Yes	Yes
Expected GDP growth 2020/2021	Yes	Yes	Yes	Yes
Industry fixed effects	Yes	Yes	Yes	Yes
N	1,913	1,913	1,913	1,913

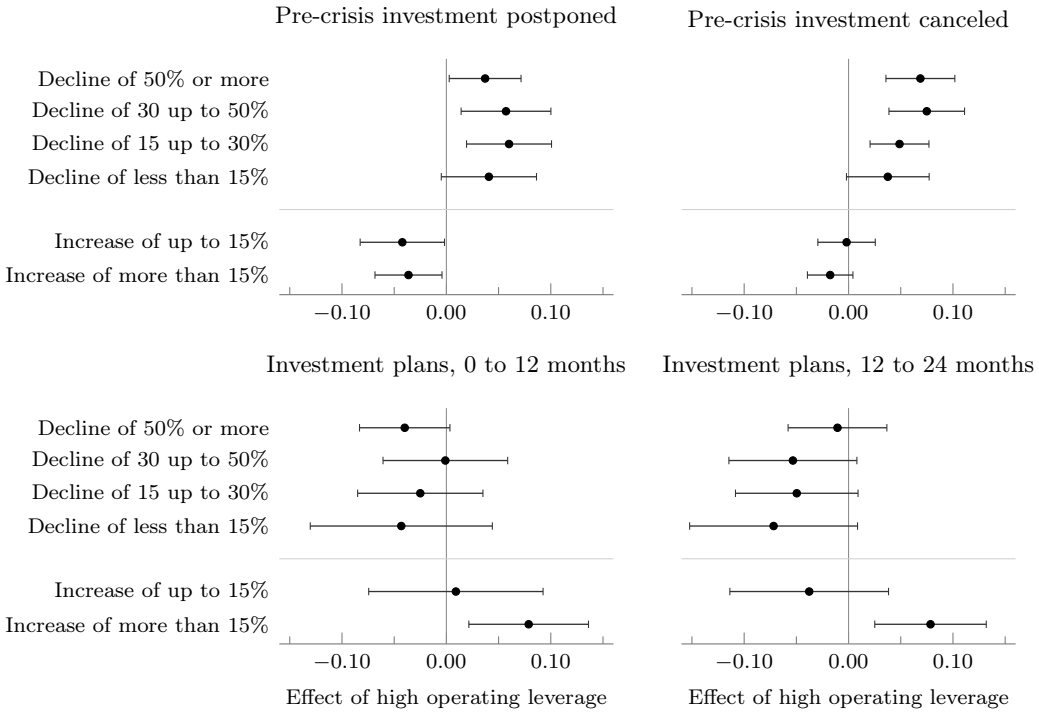
This table reports OLS estimates for the 1,913 firms that experienced a positive revenue shock. Firms are classified as having high (low) operating leverage if their profit increased by more (less) than predicted by the first-stage regression in Section 3.2.1, which is the mirror of the classification used after a revenue decline. In columns (1) and (2), the dependent variable is the share of the investment planned before the crisis that the firm postpones or cancels; in columns (3) and (4), it is an indicator for current plans to invest within the next 12 months or in 12 to 24 months. The p-value tests whether the effect differs from the effect after a revenue decline. It results from the pooled sample of all 6,786 firms, in which the classification is interacted with an indicator for a revenue decline. All regressions control for the revenue change with a separate fixed effect for each reported value, and for the revenue and employee size classes, the expected GDP growth for 2020 and 2021, and industry fixed effects. Figure 3 reports the same comparison separately by the size of the revenue change. Please refer to Table A4 for a full description of all variables. Heteroskedasticity-robust standard errors are reported in parentheses. ***, **, * indicate statistical significance at the 1%, 5%, and 10% levels (two-tailed), respectively.

The results move in the predicted direction. Firms with high operating leverage postpone 2.9 percentage points less of their planned investment and are 6.2 and 5.1 percentage points more likely to plan investment over the next twelve and the subsequent twelve to twenty-four months, compared with means of 13.7, 54.8, and 65.9 percent among firms with low operating leverage. All three estimates are significant at the five percent level. The canceled share likewise reverses sign, at −1.2 percentage points, but is not statistically significant. Cancellation is uncommon following a revenue increase: 81 percent of firms cancel none of their planned investment, and the mean canceled share among firms with low operating leverage is 6.8 percent. In relative terms, however, the estimate is similar to that for postponements, at −17 versus −21 percent. All four estimates differ from their counterparts following a revenue decline at the one percent level.

Figure 3 examines this reversal across the distribution of revenue changes. We estimate the association separately within six groups of firms defined by the size of their revenue change.[13] For the two revision shares the estimates are positive in all four groups whose revenue declined, between 3.7 and 6.0 percentage points on the postponed share and between 3.8 and 7.5 percentage points on the canceled share, and turn negative or close to zero in the two groups whose revenue increased.

[13]The decline side is split more finely because of fewer firms experiencing a revenue increase.

Figure 3: Operating Leverage and Investment, by the Revenue Shock



This figure reports the effect of high operating leverage on investment revisions and investment plans, estimated separately within groups of firms defined by their revenue change. A firm is classified as having high operating leverage if its profit moved further than predicted by the first-stage regression in Section 3.2.1: downwards by more than predicted after a revenue decline, upwards by more than predicted after a revenue increase. The first cut point is the same in both directions, at 15 percent. For larger changes, the decline side is split more finely because 4,873 firms reported a revenue decline against 1,913 reporting an increase. The resulting groups hold 1,872, 1,110, 1,176, 715, 701 and 1,212 firms, from the largest decline to the largest increase. Each estimate comes from a separate regression with the controls of Table 4, except that the revenue change enters linearly rather than through a fixed effect for each reported value, because the groups are themselves defined by the size of the revenue change. The bars are 95 percent confidence intervals from heteroskedasticity-robust standard errors.

The two investment-plan indicators are imprecisely estimated within the declining groups, but among firms whose revenue rose by more than 15 percent both are positive, at 7.9 percentage points on each horizon.

Taken together, these results are consistent with our measure capturing differences in how strongly profits respond to changes in revenue. Higher operating leverage is associated with larger reductions in investment when revenues fall but with less retrenchment when revenues rise. This reversal is difficult to explain solely by differences in firms' exposure to the crisis and is also less naturally attributed to financial constraints among firms experiencing positive revenue shocks.

5 Conclusion

In this paper, we show that operating leverage shapes how firms adjust investment following a large negative demand shock. Using survey data from 4,873 German firms that experienced a revenue

decline during the COVID-19 pandemic, we classify firms as having high operating leverage when their profits decline by more than would be expected given the revenue change and the responses of firms in the same industry and size class. Firms with high operating leverage postpone 4.8 and cancel 5.9 percentage points more of the investment they had planned before the crisis. Relative to firms with low operating leverage, these differences correspond to 19.6 and 41.3 percent of the respective shares revised. Firms with high operating leverage are also less likely to plan investment over the following two years.

Our approach also provides a way to measure operating leverage when detailed information on firms' cost structures is unavailable. Conventional measures rely either on estimating the sensitivity of operating income to sales over time (Mandelker and Rhee, 1984; O'Brien and Vanderheiden, 1987) or on accounting ratios constructed from income-statement information (Lev, 1974; Novy-Marx, 2011; Aboody et al., 2018). These approaches are difficult to apply to privately held firms for which sufficiently detailed financial statements or long time series are often unavailable. Our measure instead uses firms' reported changes in revenue and profit and benchmarks the resulting pass-through against that of comparable firms facing the same revenue shock. The follow-up survey provides direct validation of this classification, as firms identified as having higher operating leverage subsequently report greater difficulty reducing their costs in the short run. This approach may therefore be useful in settings in which cost structures cannot be observed directly.

The results also speak to why operating leverage affects investment. Prior work emphasizes that operating leverage transmits revenue shocks into profits and thereby affects risk, liquidity, and financing (Mandelker and Rhee, 1984; Chen et al., 2019). Our evidence suggests that financing conditions alone do not account for the investment response we document. In our model, past cost commitments change the attractiveness of subsequent investment. A firm that has already committed to a relatively rigid cost structure has less to gain from further cost-reducing investment, and a decline in demand reduces the return to projects whose benefits depend on the volume over which they are applied. Consistent with this mechanism, the association between operating leverage and investment revisions does not become stronger among firms for which financing constraints are more likely to bind. We also find suggestive evidence of the opposite pattern among the 1,913 firms whose revenues increased, consistent with operating leverage magnifying firms' responses to revenue shocks in both directions.

These findings have implications for how crisis policies affect investment. Liquidity support can help firms meet short-term obligations and remain in operation, but it need not preserve investment

plans if a decline in demand has reduced the expected return to the underlying projects. Our results therefore suggest that policies aimed at easing financial constraints and policies aimed at sustaining investment need not operate through the same mechanism. Support that helps a firm survive may preserve its ability to invest, while leaving unchanged its incentive to undertake projects whose returns have fallen with demand.

Our analysis is subject to several limitations. First, the model focuses on investments that reduce costs, whereas the survey records revisions to firms' investment plans without distinguishing their purpose. The mechanism we set out in the theoretical model might not apply equally to all forms of investment. It should be most relevant when the return to an investment scales with the volume over which it is used and when prior commitments have already captured part of that return. Capacity investment is one example of an investment for which these conditions are likely to hold. By contrast, investments such as R&D, product development, or market entry may depend less directly on current production volume, and our model therefore has no clear prediction for them. Distinguishing these investment types would provide a useful avenue for future research. Second, our evidence comes from a single setting. We study one unusually large and unexpected demand shock among predominantly privately held small and medium-sized firms in Germany, during a period in which firms had access to extensive government support. These features make the setting particularly useful for observing sharp revisions to investment plans, but they may also limit the extent to which the estimates generalize to smaller shocks, other institutional environments, or publicly listed firms. Future work could examine whether the same relation between operating leverage and investment appears outside crisis periods and whether it varies with firms' access to financing, competitive environment, or type of investment.

More broadly, our findings suggest that operating leverage matters not only because it affects the sensitivity of profits to revenue shocks, but also because it shapes firms' subsequent investment responses. Cost structures established before a shock affect which investments remain worthwhile once demand changes, creating a direct link between firms' prior cost commitments and their subsequent investment decisions.

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Appendix

A Proofs

Demand conditions. Let $\Lambda \equiv (d-2c_L)^2/(8d)$, $A \equiv 3\Lambda/2$, and $B \equiv \frac{1}{4}\big(3 - A/\sqrt{A^2-8}\big)$. Proposition 1 and the results that build on it assume

$$\Lambda > c_F + 1/c_F, \tag{13a}$$

$$d + 2c_L \geq 2c_F + 4/c_F, \tag{13b}$$

$$2(d-A)/(A^2-8)^{3/2} < B, \tag{13c}$$

$$1/\Lambda^2 < B. \tag{13d}$$

Condition (13a) implies $\Lambda > 2$ and hence $A > 3$, so B is well defined and positive. All four conditions require demand to be large relative to unit variable costs. They hold, for example, for all $d \geq 35$ whenever $1/2 \leq c_F \leq 1$.[14]

Proof of Lemma 1. Fix $q > 0$ and let $g(I) \equiv c_i e^{-I} q + I$. Then $g'(I) = 1 - c_i q\, e^{-I}$ and $g''(I) = c_i q\, e^{-I} > 0$, so g is strictly convex on $[0,\infty)$ and has a unique minimizer. If $c_i q \leq 1$, then $g'(I) \geq 1 - e^{-I} \geq 0$ for all $I \geq 0$, so $I_i^* = 0$ and $C_i(q) = c_i q$. If $c_i q > 1$, then $g'(0) < 0$ and $g'(I) = 0$ at $I = \ln(c_i q) > 0$, which yields $C_i(q) = 1 + \ln(c_i q)$. For $q = 0$, $I_i^* = 0$ and $C_i(0) = 0$. This proves (a) and the expression in (b).

The derivative of C_i equals c_i for $q < 1/c_i$ and $1/q$ for $q > 1/c_i$. Both expressions equal c_i at $q = 1/c_i$, so C_i is continuously differentiable with $C_i'(q) = \min\{c_i, 1/q\} > 0$ for $q > 0$ and (right) derivative $C_i'(0) = c_i$. Because C_i' is nonincreasing, C_i is concave. Part (c) follows from differentiating $I_i^* = \ln c_i + \ln q$: $\partial I_i^*/\partial c_i = 1/c_i$ and $\partial \ln I_i^*/\partial \ln q = (1/I_i^*)\,(\partial I_i^*/\partial \ln q) = 1/I_i^*$. □

Proof of Lemma 2. By Lemma 1, the follower maximizes $\varphi(q) \equiv (a-q)q - C_F(q) - K_F$ over $q \geq 0$. The function φ is continuously differentiable with $\varphi'(q) = a - 2q - \min\{c_F, 1/q\}$ for $q > 0$ and $\varphi'(0) = a - c_F$. On $(0, 1/c_F)$, $\varphi''(q) = -2 < 0$. On $(1/c_F, \infty)$, $\varphi''(q) = -2 + 1/q^2 < 0$, because $q > 1/c_F \geq 1$ by condition (2). Hence φ' is continuous and strictly decreasing on $[0,\infty)$, so φ is strictly concave. Because $\varphi'(q) \leq a - 2q \to -\infty$ as $q \to \infty$, φ has a unique maximizer.

If $\varphi'(0) = a - c_F \leq 0$, the maximizer is $q_F^* = 0$. Otherwise it is the unique root of φ'. This root

[14] We verified this claim numerically on a fine grid of (c_L, c_F, d). Numerical solutions of the model further indicate that the characterization in Proposition 1 continues to hold for considerably lower levels of demand, for example for all $d \geq 11$ when $1/2 \leq c_F \leq 1$.

lies in $(0, 1/c_F]$ if and only if $\varphi'(1/c_F) = a - 2/c_F - c_F \leq 0$, in which case $q_F^* = (a - c_F)/2$ and, by Lemma 1(a), the follower does not invest. If $a > c_F + 2/c_F$, the root lies in $(1/c_F, \infty)$ and solves $a - 2q - 1/q = 0$, that is, $2q^2 - aq + 1 = 0$. Because $c_F + 2/c_F \geq 2\sqrt{2}$, the discriminant $a^2 - 8$ is positive. The two roots, $(a \pm \sqrt{a^2 - 8})/4$, have product $1/2$. The smaller root is therefore at most $1/\sqrt{2} < 1 \leq 1/c_F$ and is not admissible. Hence $q_F^* = (a + \sqrt{a^2 - 8})/4 > 1/c_F$, and the follower invests. This establishes (6).

At $a = c_F + 2/c_F$, we have $\sqrt{a^2 - 8} = |2/c_F - c_F| = 2/c_F - c_F$, because $c_F \leq 1 < \sqrt{2}$. Both expressions in (6) then equal $1/c_F$, and at $a = c_F$ both the first and the second expression equal zero. Hence q_F^* is continuous in a and, since $a = d - q_L$, in q_L. In the investment region, differentiating with respect to q_L (using $da/dq_L = -1$) gives (7). Because $x + 2/x$ is decreasing on $(0, \sqrt{2})$ and $c_F \leq 1$, the investment region satisfies $a > c_F + 2/c_F \geq 3$. Hence $a^2 > 9$, which is equivalent to $a/\sqrt{a^2 - 8} < 3$. Together with $a/\sqrt{a^2 - 8} > 1$, this yields the bounds in (7). In the other two regions, $dq_F^*/dq_L \in \{-1/2,\ 0\}$, so q_F^* is nonincreasing in q_L.

Finally, $p(q_L) = a - q_F^*(a)$ is continuous. Its derivative with respect to q_L equals -1 if the follower is inactive and $-1/2$ if the follower produces without investing. In the investment region, it equals

$$p'(q_L) = -\frac{1}{4}\left(3 - \frac{a}{\sqrt{a^2 - 8}}\right) < 0, \tag{14}$$

where the inequality follows from $a/\sqrt{a^2 - 8} < 3$. Hence p is continuous and piecewise strictly decreasing, and therefore strictly decreasing in q_L. As a function of a, the price $a - q_F^*(a)$ is strictly increasing, with derivative 1, $1/2$, or $(3 - a/\sqrt{a^2 - 8})/4 > 0$ in the three regions. □

Proof of Proposition 1. *Step 0 (Reduction and existence).* For any q_L, the follower's best response, and hence the price, depend only on q_L (Lemma 2). For a given q_L, the leader therefore maximizes its profit by minimizing costs, that is, by investing $I_L^*(q_L)$ (Lemma 1). This reduces the leader's problem to (8). For $q_L > d$, the price is negative and $\Pi_L(q_L) < -K_L = \Pi_L(0)$, so restricting attention to $[0, d]$ is without loss of generality. Because p and C_L are continuous, Π_L is continuous on the compact set $[0, d]$ and attains a maximum. In Steps 1–4, we write Π_L net of the constant K_L.

Step 1 (A lower bound on the leader's profit). By conditions (13b) and (2), $d \geq 2(c_F - c_L) + 4/c_F > 4 > 2c_L$. Let $\tilde{q} \equiv (d - 2c_L)/2 > 0$ and $\tilde{a} \equiv d - \tilde{q} = (d + 2c_L)/2$. By condition (13b), $\tilde{a} \geq c_F + 2/c_F$, so by Lemma 2 the price at $\tilde{q}$ is $p(\tilde{q}) = (3\tilde{a} - \sqrt{\tilde{a}^2 - 8})/4 \geq \tilde{a}/2$. (At $\tilde{a} = c_F + 2/c_F$,

both expressions in (6) coincide.) Because $C_L(q) \le c_L q$,

$$\Pi_L(\tilde{q}) \ge \tilde{q}\left(\frac{\tilde{a}}{2} - c_L\right) = \frac{(d-2c_L)^2}{8} = d\,\Lambda.$$

Step 2 (Localization). Consider any $q_L \in [0,d]$ with $q_L < \Lambda$. Because $p(q_L) \le d - q_L < d$, we have $\Pi_L(q_L) \le p(q_L)\,q_L < d\Lambda$ if $q_L > 0$, and $\Pi_L(0) = 0 < d\Lambda$. Next, consider any q_L with $p(q_L) < \Lambda$. Then $\Pi_L(q_L) \le p(q_L)\,q_L < \Lambda d$ if $p(q_L) > 0$, and $\Pi_L(q_L) \le 0$ otherwise. By Step 1, every maximizer therefore lies in

$$W \equiv \{\, q_L \in [\Lambda, d] : p(q_L) \ge \Lambda \,\}.$$

The set W contains $\tilde{q}$, because $\Lambda = \tilde{q}\,(d-2c_L)/(4d) < \tilde{q}$ and $p(\tilde{q}) \ge \tilde{a}/2 > d/4 > \Lambda$. At $q_L = d$, we have $a = 0 \le c_F$, so $q_F^* = 0$ and $p(d) = 0 < \Lambda$. Because p is continuous and strictly decreasing (Lemma 2), there is a unique $\bar{q} \in (\tilde{q}, d)$ with $p(\bar{q}) = \Lambda$, and $W = [\Lambda, \bar{q}]$ is a nonempty closed interval. The endpoints of W are not maximizers. At $q_L = \Lambda$, $\Pi_L \le (d-\Lambda)\Lambda < d\Lambda$. At $q_L = \bar{q}$, $\Pi_L \le \Lambda\bar{q} < d\Lambda$. Hence every maximizer lies in the interior of W.

Step 3 (The follower invests on W). If the follower does not invest, that is, $a \le c_F + 2/c_F$, then (6) implies $p = (a + c_F)/2 \le c_F + 1/c_F$ if the follower produces and $p = a \le c_F$ if it does not. By condition (13a), $c_F + 1/c_F < \Lambda$, so such q_L do not belong to W. Hence, for all $q_L \in W$, the follower invests, $q_F^* = (a+S)/4$ with $S = \sqrt{a^2 - 8}$, and $p = (3a - S)/4$.

Step 4 (Strict concavity on W). On W, revenue $R(q_L) = p(q_L)\,q_L$ has second derivative $R'' = 2p' + q_L\,p''$. By (14) and $d(a/S)/dq_L = 8/S^3$, we have $p'' = 2/S^3$ and

$$R''(q_L) = -\frac{1}{2}\left(3 - \frac{a}{S}\right) + \frac{2q_L}{S^3}.$$

By Lemma 1(b), $-C_L''(q_L) \in \{0,\ 1/q_L^2\}$ wherever it exists, and $1/q_L^2 \le 1/\Lambda^2$ on W. We next bound a from below. Write $s \equiv q_F^*$. The follower's first-order condition, $a = 2s + 1/s$, implies $p = a - s = s + 1/s$. Because $s > 1/c_F \ge 1$, we have $p \le 2s$ and hence $a = p + s \ge 3p/2 \ge 3\Lambda/2 = A$. Because a/S is decreasing and S is increasing in a, and because $q_L = d - a \le d - A$, we obtain for every $q_L \in W$

$$\frac{1}{4}\left(3 - \frac{a}{S}\right) \ge B, \qquad \frac{2q_L}{S^3} \le \frac{2(d-A)}{(A^2-8)^{3/2}}, \qquad \frac{1}{q_L^2} \le \frac{1}{\Lambda^2}. \tag{15}$$

Hence, by conditions (13c) and (13d),

$$\Pi_L''(q_L) \le -2B + \frac{2(d-A)}{(A^2-8)^{3/2}} + \frac{1}{\Lambda^2} < -2B + B + B = 0.$$

This holds at every $q_L \in W$ other than $q_L = 1/c_L$, where C_L'' does not exist. Because Π_L' is continuous (Lemma 1(b)) and strictly decreasing on each side of $1/c_L$, it is strictly decreasing on W. Hence Π_L is strictly concave on W and has at most one maximizer on W. By Step 0, a global maximizer exists, and by Step 2 every global maximizer lies in the interior of W. Hence the global maximizer q_L^* is unique and satisfies $\Pi_L'(q_L^*) = 0$. Using $p = (3a - S)/4$ (Step 3), p' from (14), and $C_L'(q_L) = \min\{c_L, 1/q_L\}$, this condition is

$$\frac{3a-S}{4} - q_L\,\frac{3 - a/S}{4} = \min\left\{c_L,\ \frac{1}{q_L}\right\}, \qquad a = d - q_L, \quad S = \sqrt{a^2-8}. \tag{16}$$

Step 5 (Equilibrium properties). Because the leader's maximizer is unique and the follower's best response is unique for every q_L, the subgame-perfect equilibrium is unique. Part (a): $q_L^* > \Lambda > 0$, and by Step 3 the follower produces $q_F^* > 1/c_F$ and invests $I_F^* = \ln(c_F q_F^*) > 0$. Moreover, the follower's first-order condition $a = 2q_F + 1/q_F$ (Step 3) gives $p^* = a^* - q_F^* = q_F^* + 1/q_F^*$. Part (b) follows from Lemma 1(a).

Part (c). Fix d, and consider two cost parameter vectors that satisfy conditions (2) and (13) and for which the leader invests in equilibrium. Denote the corresponding objects by $(\Lambda^{(1)}, W^{(1)}, q_L^{(1)})$ and $(\Lambda^{(2)}, W^{(2)}, q_L^{(2)})$. Define, for q_L with $a = d - q_L > 3$,

$$G(q_L) \equiv \frac{3a-S}{4} - q_L\,\frac{3 - a/S}{4} - \frac{1}{q_L}, \qquad S = \sqrt{a^2-8},$$

which does not depend on any cost parameter. Because the leader invests, the right-hand side of (16) equals $1/q_L$ at the respective equilibrium, so $G(q_L^{(1)}) = G(q_L^{(2)}) = 0$. Moreover, $G'(q_L) = R''(q_L) + 1/q_L^2$, so by the bounds in Step 4, $G' < 0$ on $W^{(1)}$ and on $W^{(2)}$.

Next, we show that $W^{(1)} = [\Lambda^{(1)}, \bar{q}^{(1)}]$ and $W^{(2)} = [\Lambda^{(2)}, \bar{q}^{(2)}]$ are nested. By Step 3, the follower invests at every point of $W^{(1)}$ and $W^{(2)}$. The price on these sets is therefore given by the same function $\hat{p}(q_L) \equiv (3a - S)/4$, which is strictly decreasing wherever $a > 3$ (by (14)). The upper endpoints satisfy $\hat{p}(\bar{q}^{(k)}) = \Lambda^{(k)}$, $k = 1, 2$ (Step 2). If, say, $\Lambda^{(1)} \le \Lambda^{(2)}$, then $\bar{q}^{(1)} \ge \bar{q}^{(2)}$ and hence $W^{(2)} \subseteq W^{(1)}$. Both $q_L^{(1)}$ and $q_L^{(2)}$ then lie in $W^{(1)}$, on which G is strictly decreasing. Because G has at most one zero on $W^{(1)}$, $q_L^{(1)} = q_L^{(2)}$. The case $\Lambda^{(1)} > \Lambda^{(2)}$ is symmetric. Finally, $q_F^* = (a^* + S^*)/4$ and $p^* = q_F^* + 1/q_F^*$ depend only on the leader's output and d. □

Proof of Lemma 3. Fix the cost parameters and write $\Pi_L(q;t)$, $p(q;t)$, $W(t)$, $\Lambda(t)$, and $q^*(t) \equiv q_L^*(t)$ to make the dependence on demand $t \in [d_1, d_0]$ explicit. By Proposition 1, $q^*(t)$ is the unique zero of $\Pi_L'(\cdot\,;t) \equiv \partial\Pi_L/\partial q$ on $W(t)$, and it lies in the interior of $W(t)$. Because $p(q;t)$ is continuous in (q,t) and strictly decreasing in q, and $\Lambda(t)$ is continuous, the endpoints of $W(t)$ depend continuously on t.

Step 1 (Local monotonicity of q^*). Fix $t \in [d_1, d_0]$ and $q \equiv q^*(t)$. For q in the region in which the follower invests, $\Pi_L'(q;t) = p + q\,p_q - C_L'(q)$ with $p = (3a - S)/4$, $p_q = -(3 - a/S)/4$, and $a = t - q$. Differentiating with respect to t at fixed q gives

$$\frac{\partial \Pi_L'(q;t)}{\partial t} = \frac{1}{4}\left(3 - \frac{a}{S}\right) - \frac{2q}{S^3},$$

which does not involve C_L. At any demand level s with $q \in W(s)$, the bounds (15) (with A and B evaluated at s) and condition (13c) at s imply that this derivative is at least $B - 2(s - A)/(A^2 - 8)^{3/2} > 0$. Because q lies in the interior of $W(t)$ and $W(\cdot)$ is continuous, there is $\varepsilon > 0$ such that $q \in W(s)$ for all $s \in (t - \varepsilon, t + \varepsilon)$. For $t' \in (t, t + \varepsilon)$, integrating the derivative over $s \in [t, t']$ gives $\Pi_L'(q;t') > \Pi_L'(q;t) = 0$. Because $\Pi_L'(\cdot\,;t')$ is strictly decreasing on $W(t')$ and vanishes at $q^*(t')$, it follows that $q^*(t') > q$. Analogously, $q^*(t') < q$ for $t' < t$ close to t.

Step 2 (Local monotonicity of a^*). Let $a^*(t) \equiv t - q^*(t)$. Fix t, $q \equiv q^*(t)$, and $a \equiv a^*(t)$, and consider $t' = t + \Delta$ with $\Delta > 0$ small and $q' \equiv q + \Delta$, so that $t' - q' = a$. Because p and p_q depend on (q,t) only through a, we have

$$\Pi_L'(q';t') - \Pi_L'(q;t) = \Delta\, p_q(a) - \big[C_L'(q') - C_L'(q)\big] \le -\Delta\,\frac{1}{4}\left(3 - \frac{a}{S}\right) + \left(\frac{1}{q} - \frac{1}{q'}\right) < \Delta\left(-B + \frac{1}{\Lambda(t)^2}\right) < 0,$$

where we used that $C_L'(x) = \min\{c_L, 1/x\}$ satisfies $C_L'(q) - C_L'(q') \le 1/q - 1/q' = \Delta/(qq') < \Delta/q^2 \le \Delta/\Lambda(t)^2$, the bounds (15), and condition (13d). For Δ small, q' lies in $W(t')$, so $\Pi_L'(q';t') < 0$ implies $q^*(t') < q'$, that is, $a^*(t') > a$. Analogously, using that $B > 1/\Lambda(t)^2 \ge 1/q^2$ holds with strict inequality and therefore also for nearby values, $a^*(t') < a$ for $t' < t$ close to t.

Step 3 (Global monotonicity). A function f on an interval with the property that, for every t, $f(t') > f(t)$ for all $t' > t$ close to t and $f(t') < f(t)$ for all $t' < t$ close to t, is strictly increasing.[15] By Steps 1 and 2, q^* and a^* are strictly increasing on $[d_1, d_0]$. The latter implies $q_L^*(d_0) - q_L^*(d_1) < d_0 - d_1$. Because q_F^* is strictly increasing in a in the region in which the follower

[15] Let $t < t''$ and $\tau \equiv \sup\{s \in [t, t''] : f(r) > f(t) \text{ for all } r \in (t, s]\}$. The local property at t implies $\tau > t$. By the local property at τ, there is $r \in (t, \tau)$ with $f(t) < f(r) < f(\tau)$, so $f(\tau) > f(t)$, and $f > f(\tau)$ immediately to the right of τ. If $\tau < t''$, then $f > f(t)$ on $(t, \tau + \delta]$ for some $\delta > 0$, which contradicts the definition of τ. Hence $\tau = t''$ and $f(t'') > f(t)$.

invests (Lemma 2), and the follower invests in equilibrium (Proposition 1(a)), $q_F^*(d) = q_F^*(a^*(d))$ is strictly increasing in d.

Step 4 (Profits). For $t < t'$, the leader's equilibrium profit satisfies $\Pi_L(q^*(t');t') \geq \Pi_L(q^*(t);t') > \Pi_L(q^*(t);t)$. The strict inequality holds because $q^*(t) > 0$ and the price at a given q_L is strictly increasing in $a = t - q_L$ (Lemma 2). The follower's equilibrium profit equals $\max_q\{(a-q)q - C_F(q)\} - K_F$ evaluated at $a = a^*(t)$. This maximum is strictly increasing in a because the follower produces a positive output, and a^* is strictly increasing by Step 3. □

A cost comparison. The proof of Proposition 2(c) uses the following property. Let $c_A < c_B$. Substituting $J = I - \ln c_i$ in (5) gives $C_i(q) - \ln c_i = \min_{J \geq -\ln c_i}\{e^{-J}q + J\}$. The constraint set for c_A is contained in that for c_B, because $-\ln c_A > -\ln c_B$. Hence

$$C_A(q) - \ln c_A \geq C_B(q) - \ln c_B \quad \text{for all } q \geq 0, \tag{17}$$

with equality if and only if the unconstrained minimizer, $J = \ln q$, satisfies $\ln q \geq -\ln c_A$, that is, if and only if $c_A q \geq 1$. (For $0 < c_A q < 1$, the minimum over the smaller set is strictly larger, because the objective is strictly convex in J. For $q = 0$, the left side equals $-\ln c_A$ and the right side equals $-\ln c_B < -\ln c_A$.)

Proof of Proposition 2. *Part (a).* By Lemma 3, $q_L^1 < q_L^0$ and $q_F^1 < q_F^0$, so $x_L, x_F > 0$. By Proposition 1(a), the follower invests at every $d \in [d_1, d_0]$, so $I_F^t = \ln(c_F q_F^t)$ for $t \in \{0,1\}$, $I_F^1 < I_F^0$, and $\sigma_F = (I_F^0 - I_F^1)/I_F^0 = x_F/\ln(c_F q_F^0)$. Because $I_F^1 > 0$, $\sigma_F < 1$. The leader invests at d_0, so $I_L^0 = \ln(c_L q_L^0) > 0$. If it invests at d_1, then $I_L^1 = \ln(c_L q_L^1) < I_L^0$. Otherwise, $I_L^1 = 0 < I_L^0$.

Part (b). If the leader invests at d_1, then $\sigma_L = (I_L^0 - I_L^1)/I_L^0 = x_L/\ln(c_L q_L^0)$. By Proposition 1(c), applied at d_0 and at d_1, the outputs q_L^t and q_F^t are the same for all cost parameters that satisfy conditions (2) and (13) and for which the leader invests at both demand levels. Hence x_L, x_F, θ, q_L^0, and q_F^0 do not vary with c_L within this set, and σ_L is strictly decreasing in c_L on it. Because $x_L, x_F, I_L^0, I_F^0 > 0$,

$$\sigma_L > \sigma_F \iff \frac{x_L}{I_L^0} > \frac{x_F}{I_F^0} \iff I_L^0 < \theta\, I_F^0 \iff \ln(c_L q_L^0) < \theta \ln(c_F q_F^0) \iff c_L < \frac{(c_F q_F^0)^\theta}{q_L^0}.$$

Part (c). Denote the leader with unit variable cost c_L by A and the reference leader with $c_L' \geq c_L$ by B. Both markets have the same follower, so the leader's revenue function $R(q) \equiv p(q)\,q$ at d_1 is the same in both. Let $\Pi_k(q) \equiv R(q) - C_k(q)$, $k \in \{A, B\}$, denote the leaders' objectives at

d_1 net of committed costs. If $c'_L = c_L$, the claim follows from Proposition 1(b). Let $c'_L > c_L$.

Suppose first that $c_L \hat{q}^1_L > 1$. By (17), $\Pi_A(q) + \ln c_L \leq \Pi_B(q) + \ln c'_L$ for all q, with equality for $q \geq 1/c_L$. Because $\hat{q}^1_L > 1/c_L$ maximizes Π_B,

$$\Pi_A(\hat{q}^1_L) + \ln c_L = \Pi_B(\hat{q}^1_L) + \ln c'_L \geq \Pi_B(q) + \ln c'_L \geq \Pi_A(q) + \ln c_L \qquad \text{for all } q.$$

Hence $\hat{q}^1_L$ maximizes Π_A, and by the uniqueness in Proposition 1, leader A produces $\hat{q}^1_L$ and invests.

Suppose next that $c_L \hat{q}^1_L \leq 1$, and assume to the contrary that leader A invests at d_1. Then its output q_A satisfies $c_L q_A > 1$ and $G(q_A) = 0$, with G as defined in the proof of Proposition 1(c), and $q_A \in W_A$. Because $\Lambda = (d_1 - 2c)^2/(8d_1)$ is decreasing in c for $c < d_1/2$ and the price function is the same in both markets, $\Lambda_A > \Lambda_B$ and hence $W_A \subseteq W_B$. Leader B's output also satisfies $G(\hat{q}^1_L) = 0$ with $\hat{q}^1_L \in W_B$. Because G is strictly decreasing on W_B, $q_A = \hat{q}^1_L$, which contradicts $c_L q_A > 1 \geq c_L \hat{q}^1_L$. Hence leader A does not invest, $I^1_L = 0$, and $\sigma_L = 1 > \sigma_F$ by part (a).

Finally, let q^0_L, q^0_F, x_L, x_F, and θ denote the outputs and output declines in the market with leader B, in which the leader invests at both demand levels, so that $x_L = \ln(q^0_L/\hat{q}^1_L)$. Then $\bar{c}\,\hat{q}^1_L > 1$ is equivalent to $\theta \ln(c_F q^0_F) > x_L$, that is, to $\ln(c_F q^0_F) > x_F$, which holds because the follower's revision share in that market is below one by part (a). Hence $1/\hat{q}^1_L < \bar{c}$. □

B Supplementary Empirical Analyses

Table A1: Sample Selection

	Excluded (1)	Remaining (2)
Panel A: First survey wave (July to October 2020)		
Respondents		19,831
Less: no revenue change reported	(7,743)	12,088
Less: revenue change reported as zero	(2,091)	9,997
Less: missing responses on government support	(552)	9,445
Less: missing responses on the crisis outcomes	(2,170)	7,275
Less: missing investment and hiring plans	(9)	7,266
Less: missing change in employees or liabilities	(471)	6,795
Less: implausible firm characteristics	(9)	6,786
Thereof: negative revenue shock		4,873
Thereof: positive revenue shock		1,913
Panel B: Second survey wave (November 2020 to June 2021)		
Respondents		15,618
Less: no revenue change reported	(5,387)	10,231
Less: revenue change reported as zero	(1,859)	8,372
Thereof: negative revenue shock		5,885
Thereof: positive revenue shock		2,487
Panel firms		1,634

This table documents the selection of the estimation samples. Firms enter if they report a revenue change relative to January 2020. Firms reporting an unchanged revenue are excluded because the classification of operating leverage compares the profit change to the change predicted from the revenue change. In the first wave, firms are additionally required to report the change in their number of employees and in their liabilities and to answer the questions on government support, on the crisis outcomes, and on their investment and hiring plans, and firms whose reported size or legal form is implausible against their Orbis record are excluded. Partial responses are kept as long as the questions used are answered. The first-wave sample is constant across all outcomes as it requires answers on the crisis outcomes and on the investment and hiring plans; the second-wave sample is defined by the classification alone, so that the number of observations varies with the question and is reported for each column. The negative revenue shock samples are the basis for Tables 2 to 7; firms with a positive revenue shock are reported in Table 8. Panel firms are those that participated in both waves and reported a revenue decline in the first wave. They are used in Table 3.

Table A2: Sample Composition and Exposure to the Crisis

	Sample		Revenue change		
	N	%	Share with a decline (%)	Mean (%)	Median (%)
	(1)	(2)	(3)	(4)	(5)
Panel A: Industry (WZ 2008 sections)					
A Agriculture, forestry and fishing	52	0.8	57.7	−0.7	−6.0
B Mining and quarrying	14	0.2	57.1	−3.9	−8.5
C Manufacturing	877	12.9	78.1	−22.4	−25.0
D Electricity and gas supply	73	1.1	53.4	−2.1	−5.0
E Water supply and waste management	32	0.5	62.5	−10.1	−6.5
F Construction	376	5.5	53.5	−1.9	−2.0
G Wholesale and retail trade	1,138	16.8	64.1	−11.6	−20.0
H Transportation and storage	167	2.5	83.8	−32.5	−31.0
I Accommodation and food services	180	2.7	87.8	−55.5	−63.5
J Information and communication	885	13.0	67.8	−16.4	−20.0
K Financial and insurance activities	232	3.4	67.2	−9.9	−10.5
L Real estate activities	177	2.6	70.6	−12.1	−10.0
M Professional, scientific and technical activities	792	11.7	75.3	−26.4	−25.0
N Administrative and support services	521	7.7	75.4	−35.1	−41.0
O Public administration	25	0.4	68.0	−9.0	−15.0
P Education	95	1.4	82.1	−29.1	−34.0
Q Human health and social work	323	4.8	71.5	−13.4	−17.0
R Arts, entertainment and recreation	273	4.0	90.5	−59.9	−76.0
S Other service activities	295	4.3	77.6	−27.5	−30.0
T Households as employers	30	0.4	50.0	5.6	−3.0
U Extraterritorial organisations	7	0.1	100.0	−38.4	−40.0
Not reported	222	3.3	76.1	−31.2	−42.0
Panel B: Number of employees					
0–9	4,300	63.4	72.9	−24.8	−25.0
10–49	1,780	26.2	69.1	−16.5	−20.0
50–249	450	6.6	71.3	−12.3	−18.0
250 or more	166	2.4	72.3	−13.6	−15.0
Not reported	90	1.3	76.7	−30.4	−40.0
Panel C: Revenue in 2019 (EUR million)					
Less than 2	4,997	73.6	72.5	−23.6	−24.0
2–10	1,168	17.2	69.9	−17.2	−20.0
10–50	392	5.8	68.9	−11.8	−18.0
More than 50	162	2.4	71.0	−11.8	−12.5
Not reported	67	1.0	73.1	−28.8	−33.0
Panel D: Legal form					
Sole proprietorship	764	11.3	72.3	−24.7	−25.0
Corporation	4,828	71.1	71.3	−21.1	−21.0
Partnership	984	14.5	72.1	−21.2	−20.0
Other legal forms	205	3.0	80.5	−25.0	−21.0
Not reported	5	0.1	80.0	−10.8	−20.0
Total	6,786	100.0	71.8	−21.6	−21.0

This table reports the composition of the first-wave sample and how strongly each group was affected by the crisis. Column (1) reports the number of firms and column (2) their share of the sample. Column (3) reports the share of firms in the group that report a revenue decline relative to January 2020 and that therefore enter the samples used in Tables 2 to 7, and columns (4) and (5) the mean and the median revenue change in the group, including firms whose revenue increased. Industries are the sections of the German classification of economic activities (WZ 2008) as reported by the firm in the questionnaire. Firms that do not provide an industry are reported separately. The number of employees refers to full-time employees subject to social security. Corporations comprise the GmbH, UG, AG, KGaA, S.E., and Limited; partnerships comprise the GmbH & Co. KG, oHG, GbR, PartG, and KG; cooperatives, registered associations, public-law entities, foundations, and firms reporting another legal form are collected under other legal forms. The coverage of the German Business Panel relative to the German Statistical Business Register is documented in Bischof et al. (2025). Percentages may not sum to 100 because of rounding.

Table A3: The Respondents

Position of the respondent	Sample		Mean change (%)		Share of firms (%)	
	N	%	Revenue	Profit	Revenue change rounded to ten	High operating leverage
	(1)	(2)	(3)	(4)	(5)	(6)
Owner or managing director	3,434	70.5	−43.9	−38.2	45.3	49.4
Head of department	119	2.4	−37.4	−33.4	49.6	39.5
Clerical staff	45	0.9	−44.6	−34.8	53.3	46.7
Other position	105	2.2	−40.0	−32.1	49.5	41.0
Not reported	1,170	24.0	−43.9	−37.7	44.5	49.7
Total	4,873	100.0	−43.6	−37.8	45.4	49.1

This table describes the individuals who completed the first-wave questionnaire for the 4,873 firms with a negative revenue shock, the sample used in Tables 2 to 7. The first row covers owners, managing directors, board members and authorized signatories. Columns (3) and (4) report the mean revenue and profit changes. Column (5) is the share of firms whose reported revenue change is a multiple of ten, a measure of how precisely the figure is given. Column (6) is the share classified as having high operating leverage. It does not vary materially with the position, as a linear probability model of the classification on indicators for the five groups with the controls of the investment regressions confirms (joint test $F = 0.85$, $p = 0.49$).

Table A4: Variable Definitions

Variable	Type	Source	Definition
Panel A: Firm characteristics			
Revenue 2019 (EUR m)	Size class	Waves 1, 2	Annual revenue in 2019 in fourteen size classes. Descriptive statistics use class midpoints, with the top class set to its lower bound; the regressions use fixed effects for the classes, also interacted with the revenue change.
No. of employees	Size class	Waves 1, 2	Full-time employees subject to social security in nine size classes, used as above.
Manufacturing, Trade, Services, Other industries	Indicator	Waves 1, 2	Industry classified according to WZ 2008. Manufacturing covers divisions 10 to 33, trade divisions 45 to 47, services divisions 49 to 99, and other industries the remaining divisions. The regressions use fixed effects for the two-digit division. Firms select their industry themselves, at the level of detail they choose; the 222 firms that give no industry form their own category and are kept in the estimation.
Legal form	Class	Wave 1	Legal form reported by the firm in eighteen categories, grouped in Table A2.
Panel B: Crisis exposure and expectations			
Δ Revenue (%)	Percent	Waves 1, 2	Change of monthly revenue since the beginning of the crisis, as reported by the firm. A value below zero defines the negative revenue shock sample. Enters every regression linearly and interacted with the revenue size class.
Δ Profit (%)	Percent	Waves 1, 2	Change of monthly net income, reported in the same block of items. Input to the classification of operating leverage. Firms that report a revenue change but leave the profit change blank are set to zero; Table A7 reports the results without any firm reporting no change in profit.
Δ Employees (%)	Percent	Wave 1	Change of the number of employees, from the same block of items.
Δ Liquidity (%)	Percent	Wave 1	Change of liquidity, from the same block of items.
Δ Liabilities (%)	Percent	Wave 1	Change of liabilities, from the same block of items.
Government aid received	Indicator	Wave 1	Equals one if the firm claims at least one government support measure because of the crisis, from two lists of measures that the firm answers separately. The non-tax measures include the emergency relief, interim aid, the short-time work allowance, guarantees and the KfW programmes; the tax measures include the refund of tax prepayments for 2020, an immediate loss carryback and the deferral of tax payments. The option to be taxed as a corporation is offered only to partnerships and sole proprietorships.

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Table A4 continued

Variable	Type	Source	Definition
Aid, would (not) have survived	Indicator	Wave 1	Among firms that claim support, whether the firm states that it would have survived the crisis without it. Asked only of firms that claim at least one measure, and only in the first wave. Table 6 compares firms without aid, firms with aid that would have survived, and firms with aid that would not.
Expected GDP growth 2020 (%)	Percent	Wave 1	Expected change of German gross domestic product in 2020 relative to 2019. Firms that do not answer are set to zero and flagged with an indicator that enters every regression.
Expected GDP growth 2021 (%)	Percent	Waves 1, 2	Expected change in 2021 relative to 2020, measured as above. The second wave asks about 2021 and 2022, and the regressions on second-wave data control for both years.
Panel C: Measure of operating leverage			
High operating leverage	Indicator	Constructed	Equals one if the firm's profit fell by more than predicted from its revenue change, industry and size, and zero if it fell by less. The prediction comes from the first-stage regression in Section 3.2.1. Firms with a revenue increase are classified by the mirror rule, on whether their profit rose by more or by less than predicted, which is the classification used in Table 8 and in Figures 1 and 3. Firms are classified separately in each wave from that wave's answers.
Terciles of the degree of operating leverage	Indicator	Constructed	Firms with high operating leverage, split into three equally sized groups by the distance between their profit change and the predicted profit change, from the lowest to the highest degree. Used in Table A6.
Operating leverage (standardized)	Continuous	Constructed	Distance between the predicted and the realized profit change, standardized to mean zero and unit standard deviation and increasing in operating leverage. Used in Panel B of Table 4 and Panel B of Table 7.
Alternative degree measure	Continuous	Constructed	Ratio of the reported percentage change in profit to the reported percentage change in revenue, the common definition of the degree of operating leverage, formed without the first-stage regression. Firms are grouped into bands of this ratio in Table 5.
Panel D: Outcome variables, investment, employment and prices			
Pre-crisis investment postponed (share)	Share, 0–1	Waves 1, 2	Share of the investment that the firm had planned before the crisis and now postpones, reported in percent and rescaled to lie between zero and one. The firm reports one percentage for postponed and one for canceled investment.
Pre-crisis investment canceled (share)	Share, 0–1	Waves 1, 2	Share of the investment that the firm had planned before the crisis and now cancels altogether, measured as above.
Investment plans, 0–12 months	Indicator	Waves 1, 2	Equals one if the firm currently plans to invest within the next twelve months.

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Table A4 continued

Variable	Type	Source	Definition
Investment plans, 12–24 months	Indicator	Waves 1, 2	Equals one if the firm currently plans to invest in twelve to twenty-four months.
Hiring plans, 0–12 months	Indicator	Wave 1	Equals one if the firm currently plans to hire additional employees within the next twelve months.
Hiring plans, 12–24 months	Indicator	Wave 1	Equals one if the firm currently plans to hire additional employees in twelve to twenty-four months.
Layoffs	Indicator	Wave 1	Equals one if the firm names a reduction of its number of employees. The firm names the measures it takes in the short term to cope with the burden of the crisis, choosing from eleven measures, so that a zero means the firm does not name the measure.
Price increases	Indicator	Wave 1	Equals one if the firm names a price increase, from the same list of measures.
Panel E: Outcome variables, cost flexibility			
Difficulty of cost reduction (0–10)	Scale, 0–10	Wave 2	Stated difficulty of reducing costs within three months when demand falls, from 0 (very easy) to 10 (very difficult).
Long-term commitments	Indicator	Wave 2	Equals one if the firm names other long-term commitments among the biggest hurdles to reducing costs. The firm may name several of the seven hurdles below.
Asset disposal	Indicator	Wave 2	Equals one if the firm names that disposing of operating assets is less economical than reducing capacity or shutting down, from the same list of hurdles.
Know-how retention	Indicator	Wave 2	Equals one if the firm names the retention of know-how, from the same list of hurdles.
Planning uncertainty	Indicator	Wave 2	Equals one if the firm names planning uncertainty, from the same list of hurdles.
Legal obligations	Indicator	Wave 2	Equals one if the firm names legal obligations, from the same list of hurdles.
Employee contracts	Indicator	Wave 2	Equals one if the firm names long-term employee contracts, from the same list of hurdles.
No hurdles	Indicator	Wave 2	Equals one if the firm names no hurdle. Firms may also name other reasons, which are not reported separately.

This table defines the variables used in the analysis. The source column gives the wave whose answers the paper uses, or the Orbis database of Bureau van Dijk; variables marked as constructed are built from the survey items as described in Section 3.2.1. Changes of the key figures refer to 31 January 2020 in the first wave and to the beginning of 2020 in the second. The first survey wave was fielded between April and July 2020, the second between November 2020 and June 2021.

Table A5: Validation Within the Second Survey Wave

	Difficulty of cost reduction (0–10)	Hurdles to cost reduction						
		Long-term commitments	Asset disposal	Planning uncertainty	Legal obligations	Know-how retention	Employee contracts	No hurdles
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
High operating leverage (wave 2)	0.636***	0.071***	0.038***	0.006	0.007	0.044***	0.003	−0.016***
	(0.069)	(0.014)	(0.010)	(0.014)	(0.012)	(0.013)	(0.012)	(0.006)
Mean of low group	6.629	0.496	0.146	0.517	0.262	0.524	0.280	0.059
R^2	0.087	0.059	0.065	0.055	0.090	0.187	0.128	0.062
N	5,747	5,700	5,700	5,700	5,700	5,700	5,700	5,700
Revenue change (wave 2) × revenue size class	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Size classes, GDP expectations, industry FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

This table repeats the comparison in Table 3 with firms classified and assessed within the second survey wave, applying the first-stage regression of Section 3.2.1 to second-wave answers. The sample comprises the 5,885 firms reporting a revenue decline in that wave. Because the classification and the assessment come from the same questionnaire and the same respondent, the estimates are not separated in time. Table 3 reports the comparison that separates them. The dependent variables and the column order are as in Table 3. The regressions control for the second-wave revenue change, linearly and interacted with the revenue size class, and for the revenue and employee size classes, the expected GDP growth, and industry fixed effects. Heteroskedasticity-robust standard errors are reported in parentheses. ***, **, * indicate statistical significance at the 1%, 5%, and 10% levels (two-tailed), respectively.

Table A6: Terciles of the Degree of Operating Leverage

	Pre-crisis investment plans		Future investment plans	
	Postponed (share) (1)	Canceled (share) (2)	0–12 months (indicator) (3)	12–24 months (indicator) (4)
Lowest tercile of operating leverage	0.026*	0.029**	−0.059***	−0.068***
	(0.014)	(0.012)	(0.020)	(0.020)
Middle tercile of operating leverage	0.053***	0.066***	−0.025	−0.031
	(0.015)	(0.014)	(0.020)	(0.021)
Highest tercile of operating leverage	0.068***	0.085***	−0.022	−0.037*
	(0.015)	(0.013)	(0.020)	(0.020)
p-value: lowest = highest tercile	0.019	0.001	0.139	0.230
p-value: all three terciles equal	0.060	0.002	0.262	0.319
Mean of low group	0.246	0.143	0.411	0.576
R^2	0.100	0.168	0.114	0.134
Revenue change fixed effects	Yes	Yes	Yes	Yes
Revenue size class	Yes	Yes	Yes	Yes
Employee size class	Yes	Yes	Yes	Yes
Expected GDP growth 2020/2021	Yes	Yes	Yes	Yes
Industry fixed effects	Yes	Yes	Yes	Yes
N	4,873	4,873	4,873	4,873

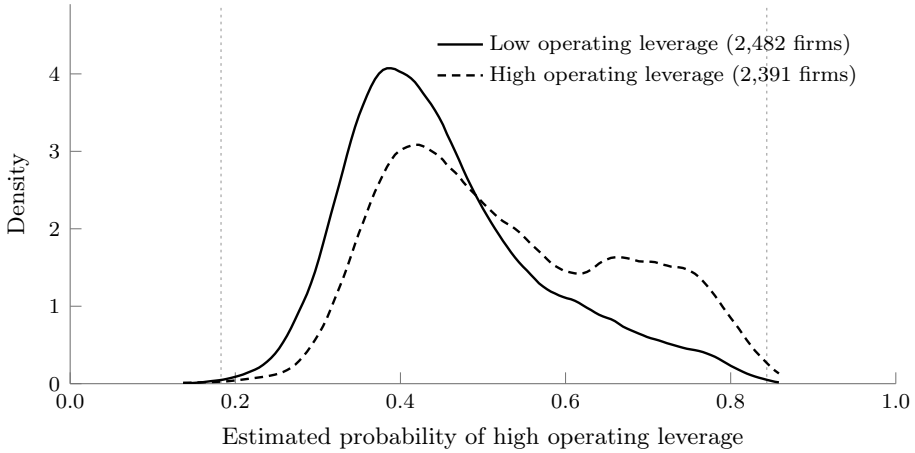
This table reports OLS estimates of investment revisions and investment plans on terciles of the degree of operating leverage for firms that experienced a negative revenue shock. Firms are classified as having high (low) operating leverage if their profit declined by more (less) than predicted by the first-stage regression in Section 3.2.1. Firms with high operating leverage are split into three groups of 797 firms each, from the lowest to the highest degree of operating leverage; the 2,482 firms with low operating leverage are the reference group. The first p-value tests whether the lowest and the highest tercile carry the same coefficient, the second whether all three do. In columns (1) and (2), the dependent variable is the share of the investment planned before the crisis that the firm postpones or cancels; in columns (3) and (4), it is an indicator for current plans to invest within the next 12 months or in 12 to 24 months. All regressions control for the revenue change with a separate fixed effect for each reported value, and for the revenue and employee size classes, the expected GDP growth for 2020 and 2021, and industry fixed effects. The table is the non-parametric counterpart of Panel B of Table 4. Please refer to Table A4 for a full description of all variables. Heteroskedasticity-robust standard errors are reported in parentheses. ***, **, * indicate statistical significance at the 1%, 5%, and 10% levels (two-tailed), respectively.

Table A7: Robustness of the Association Between Operating Leverage and Investment

	Pre-crisis investment plans				Future investment plans				
	Postponed (share)		Canceled (share)		0–12 months		12–24 months		
	Coef.	SE	Coef.	SE	Coef.	SE	Coef.	SE	N
Main specification (Table 4)	0.048^{***}	(0.010)	0.059^{***}	(0.009)	-0.036^{***}	(0.014)	-0.046^{***}	(0.014)	4,873
Panel A: Alternative classification									
Separate slopes for revenue increases and decreases	0.049^{***}	(0.010)	0.058^{***}	(0.008)	-0.039^{***}	(0.014)	-0.050^{***}	(0.014)	4,873
Quadratic revenue change in the first stage	0.049^{***}	(0.010)	0.056^{***}	(0.009)	-0.038^{***}	(0.014)	-0.050^{***}	(0.014)	4,873
Panel B: Alternative controls									
Linear control for the revenue change	0.046^{***}	(0.010)	0.058^{***}	(0.009)	-0.031^{**}	(0.014)	-0.043^{***}	(0.014)	4,873
Minimal controls	0.050^{***}	(0.010)	0.057^{***}	(0.009)	-0.034^{**}	(0.014)	-0.043^{***}	(0.014)	4,873
Controlling for rounded reporting	0.049^{***}	(0.010)	0.059^{***}	(0.009)	-0.038^{***}	(0.014)	-0.045^{***}	(0.014)	4,873
Adding the expected survival of peers	0.047^{***}	(0.010)	0.055^{***}	(0.008)	-0.034^{**}	(0.014)	-0.043^{***}	(0.014)	4,873
With crisis-period controls	0.028^{***}	(0.010)	0.041^{***}	(0.009)	-0.029^{*}	(0.015)	-0.053^{***}	(0.015)	4,832
Panel C: Alternative samples									
Excluding zero profit change	0.039^{***}	(0.011)	0.063^{***}	(0.009)	-0.038^{**}	(0.015)	-0.058^{***}	(0.015)	4,307
Excluding profit change of -100%	0.043^{***}	(0.011)	0.053^{***}	(0.009)	-0.043^{***}	(0.015)	-0.055^{***}	(0.016)	4,189
Panel D: Alternative estimators and inference									
Standard errors clustered by industry	0.048^{***}	(0.009)	0.059^{***}	(0.008)	-0.036^{**}	(0.016)	-0.046^{***}	(0.016)	4,873
Fractional logit / probit (AME)	0.047^{***}	(0.010)	0.059^{***}	(0.008)	-0.037^{***}	(0.014)	-0.045^{***}	(0.014)	4,873
Propensity score matching, 5 neighbors	0.045^{***}	(0.011)	0.064^{***}	(0.011)	-0.027^{*}	(0.015)	-0.035^{**}	(0.016)	4,873

This table reports the coefficient on the indicator for high operating leverage from variations of the main specification in Table 4, which Figure 2 plots. Unless a row states otherwise, every variation retains the separate fixed effect for each reported revenue change that Table 4 uses. Panel A reclassifies firms, re-estimating the first stage of Section 3.2.1 with separate intercepts and revenue slopes for firms whose revenue rose and fell, and with a quadratic term in the revenue change. Panel B replaces those fixed effects with a linear control interacted with the revenue size class, reduces the controls to the revenue change, the size classes and industry fixed effects, adds an indicator for the 34.4 percent of firms that report both the revenue and the profit change as a multiple of ten, adds the firm's expectation of the share of its peers that will survive, and adds the changes in employees, liquidity and liabilities together with indicators for government aid measures. Panel C excludes firms reporting a profit change of zero or of -100%. Panel D clusters standard errors by industry, estimates nonlinear models (fractional logit for the shares and probit for the indicators, reporting average marginal effects; N is 4,845 and 4,859 in the probit models), and reports the average treatment effect on the treated from propensity score matching with five neighbors and replacement. The propensity score is estimated with a probit model on the revenue change and its square, the size classes, the expected GDP growth and sector indicators (including an indicator for unknown industry). The estimated propensity scores range from 0.137 to 0.858, and 6 of the 2,391 firms with high operating leverage lie outside the range spanned by the firms with low operating leverage; Figure A1 plots the two distributions. Unless stated otherwise, standard errors are heteroskedasticity-robust. ***, **, * indicate statistical significance at the 1%, 5%, and 10% levels (two-tailed), respectively.

Figure A1: Overlap of the Propensity Score



This figure reports the distribution of the estimated propensity score in the two groups of firms with a negative revenue shock. Firms are classified as having high (low) operating leverage if their profit declined by more (less) than predicted by the first-stage regression in Section 3.2.1. The propensity score is the fitted probability from a probit model of the classification on the revenue change and its square, the revenue and employee size classes, the expected GDP growth, and indicators for manufacturing, trade and unknown industry, the same specification that underlies the matching estimates in Panel D of Appendix Table A7. Both densities use an Epanechnikov kernel with the common bandwidth 0.022. The dashed vertical lines mark the range of propensity scores spanned by the firms with low operating leverage, from 0.18 to 0.84; 6 of the 2,391 firms with high operating leverage lie outside it. The two distributions differ in location, which is what the controls adjust for, but they share almost the whole support, so the matching estimates do not rest on extrapolation.